

REALTEK

RTL9335-VT-CG

760G L3 mGig Switch with 25G/100G Uplinks

Preliminary Datasheet

(CONFIDENTIAL: Development Partners Only)

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REALTEK

Realtek Semiconductor Corp.

No. 2, Innovation Road II, Hsinchu Science Park, Hsinchu 300, Taiwan

Tel.: +886-3-578-0211 Fax: +886-3-577-6047

www.realtek.com

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USING THIS DOCUMENT

This document is intended for the software engineer's reference and provides detailed programming information.

Though every effort has been made to ensure that this document is current and accurate, more information may have become available subsequent to the production of this guide.

ELECTROSTATIC DISCHARGE (ESD) WARNING

This product can be damaged by Electrostatic Discharge (ESD). When handling, care must be taken. Damage due to inappropriate handling is not covered by warranty.

Do not open the protective conductive packaging until you have read the following, and are at an approved anti-static workstation.

- Use an approved anti-static mat to cover your work surface
- Use a conductive wrist strap attached to a good earth ground
- Always discharge yourself by touching a grounded bare metal surface or approved anti-static mat before picking up an ESD-sensitive electronic component
- If working on a prototyping board, use a soldering iron or station that is marked as ESD-safe
- Always disconnect the microcontroller from the prototyping board when it is being worked on

REVISION HISTORY

Revision	Release Date	Summary
0.1	2023/08/31	First release.
0.2	2024/01/05	Corrected typing errors
0.3	2024/02/02	RESERVED_AD35 pin change to DRAM_SEL pin
0.4	2024/10/25	Corrected typing errors

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1. General Description

The RTL9335 is a multi-layer stackable switch device with 40G/50G/100G uplinks which suitable for Enterprise/Campus aggregation, multi-gigabit access and Carrier Edge applications. With 16x10G and 24x25G SerDes embedded, a wide variety of port configurations can be supported. The maximum port count of 25G is 24, whereas 40G/100G is 4. The typical port configurations are listed below:

- 48x2.5G/5G + 4x25G + 2x100G
- 32x2.5G + 16x10G + 4x100G
- 36x2.5G + 12x10G + 4x100G
- 16x10G + 8x25G + 4x100G
- 24x10G + 4x100G
- 24x25G

The maximum core bandwidth of RTL9335 is 440Gbps, for the port configuration with I/O bandwidth larger than 440Gbps, the oversubscription function is supplied for absorbing the busty of short packets to prevent packet drop. With oversubscription function, the higher I/O bandwidth, the longer line rate packet length.

The RTL9335 comes with Realtek new generation switching architecture to support the comprehensive and emerging functions, such as EVPN-VXLAN, SRv6, TSN, Discard Monitor, Latency Monitor and so on. And all the architecture differences between previous chips and the RTL9335 are covered by Realtek SDK, to make the differences invisible to the users as possible.

The device further integrates a Quad-Core 1.2GHz ARM A55 CPU subsystem with SPI-NOR flash, SPI-NAND flash, Parallel-NAND flash, eMMC and DDR4 controller support. It also supports an out-of-band management port/NIC for management purpose. In case higher CPU performance is required, PCIe Gen3x2 interface is supplied for connecting external CPU.

2. Features

- CPU Subsystem
 - ◆ ARM A55 Quad core 1.2GHz
 - ◆ 8GB 16/32-bit DDR4, ECC support
 - ◆ 64MB SPI-NOR flash/512MB SPI-NAND flash/2GB Parallel-NAND flash
 - ◆ eMMC
 - ◆ USB host 3.0
 - ◆ PCIe Gen3x2 RC and EP
 - ◆ Watchdog
 - ◆ Secure Boot(Signature, Encryption)
 - ◆ Out-of-band management port
 - ◆ 3 * UART supported
 - ◆ 32 * GPIO pins with interrupt function
- Hardware Capacity and Interface
 - ◆ Supports 16 pairs of 10G Rate and 24 pairs of 25G Rate serdes interfaces
 - ◆ 8MB packet buffer
 - ◆ 12KB jumbo frame
- Stacking
 - ◆ Maximum 16 stacking devices
 - ◆ Chain/Ring/CLOS/Star topology
 - ◆ Cascade flow control
 - ◆ E2E congestion control
 - ◆ Trunk/Mirror across devices
 - ◆ Stacking over Ethernet
 - ◆ Two dedicated control queues
 - ◆ Two dedicated stacking trunks
- Port Extender
 - ◆ CB and PE support
- ◆ 8K CB extended port
- ◆ 4K CB ECID replication
- ◆ Maximum 112K PE ECID channel (share MAC table)
- MAC and Port Capability
 - ◆ 96K shared forwarding table (SFT)
 - ◆ Up to 112K MAC table (dedicated 16K + 96K SFT)
 - ◆ MAC notification
- Monitoring and Visibility
 - ◆ 4 mirror set
 - ◆ RSPAN/ERSPAN
 - ◆ sFlow/IPFIX
 - ◆ Buffer monitor
 - ◆ Discard monitor
 - ◆ Per flow latency monitor
- VLAN
 - ◆ 12K Ingress and 8K Egress VLAN translation
 - ◆ 12K MAC-based VLAN(share with ingress VLAN translation table)
 - ◆ N:1 VLAN translation
 - ◆ 8K virtual VLAN
- Link Aggregation
 - ◆ 128 trunk group, each with up to 8 members
 - ◆ Per LAG hash keys
 - ◆ Local first load balancing in stacking
 - ◆ Hardware auto fail-over link down ports in standalone

■ Multicast

- ◆ IGMPv2/MLDv3 ASM/SSM lookup support
- ◆ PIM-SM/PIM-DM/PIM-BIDIR lookup support
- ◆ 16K multicast port mask table

■ QoS

- ◆ 8 UC queues + 8 MC queues + 1 Mirror queue for normal port
- ◆ 10 queues for stacking port
- ◆ 48 CPU queues
- ◆ Unicast and multicast Storm control with PPS and BPS mode
- ◆ Ingress and egress port rate limit
- ◆ WRR/WERR/Strict Priority scheduling
- ◆ 3-level hierarchical QoS
- ◆ WRED
- ◆ ECN Marking
- ◆ PFC

■ Access Control List

- ◆ VACL(1.5K)/IACL(5K)/EACL(1K)
- ◆ 226 bit match key, maximum 904 bit by four rule aggregation
- ◆ Per port per block template
- ◆ 1K IACL meter/512 EACL meter
- ◆ DLB/srTCM/trTCM/modified trTCM
- ◆ 96K Full Match ACL (share with SFT)
- ◆ 8 set 2-byte user defined fields

■ Security

- ◆ 12K IP-MAC-Port binding (share with ingress VLAN translation table)
- ◆ System/Port/VLAN based MAC constraint

◆ L2~L4 DoS Prevention

- ◆ MAC white list and black hole support
- ◆ Level-based port movement control
- ◆ 802.1x support

■ Layer 3

- ◆ IPv4/IPv6 unicast and multicast routing
- ◆ 32K IPv4/16K IPv6 LPM network route
- ◆ Maximum 112K(16K+SFT) IPv4/56K(8K+SFT) IPv6 host route
- ◆ 8K L3 interface/4K VRF/32K NextHop
- ◆ 1K ECMP group, each with up to 64 members
- ◆ uRPF
- ◆ Route twice (Overlay and Underlay)
- ◆ Micro Segmentation
- ◆ NAT/NAPT

■ Tunneling

- ◆ 4K IPv4/2K IPv6 tunnel
- ◆ IP-in-IP/L2 GRE/L3 GRE/6to4/6rd/ISATAP
- ◆ VXLAN/VXLAN-GPE/NVGRE/GENEVE
- ◆ Programmable tunnel for unknown protocol

■ EVPN

- ◆ VXLAN/VXLAN-GPE
- ◆ Two-pass Forwarding process for BUD traffic
- ◆ Multipath/aliasing
- ◆ Ingress Replication (Head-end Replication)
- ◆ Designated Forwarder (DF)
- ◆ Split-Horizon Filtering

■ Carrier Ethernet

- ◆ VPWS/VPLS/L3 MPLS
- ◆ 802.3ah OAM
- ◆ 802.1ag/Y.1731 CFM
- ◆ BFD
- ◆ MEC/GTP-u offload

◆ SRv6 & u-SID

- ◆ VLAN cross connect

■ Time Sync

- ◆ TSN (802.3br/1Qbu/1CB/1CM)
- ◆ 1588v2/PTP/SyncE

3. Block Diagram

3.1. RTL9335 Block Diagram

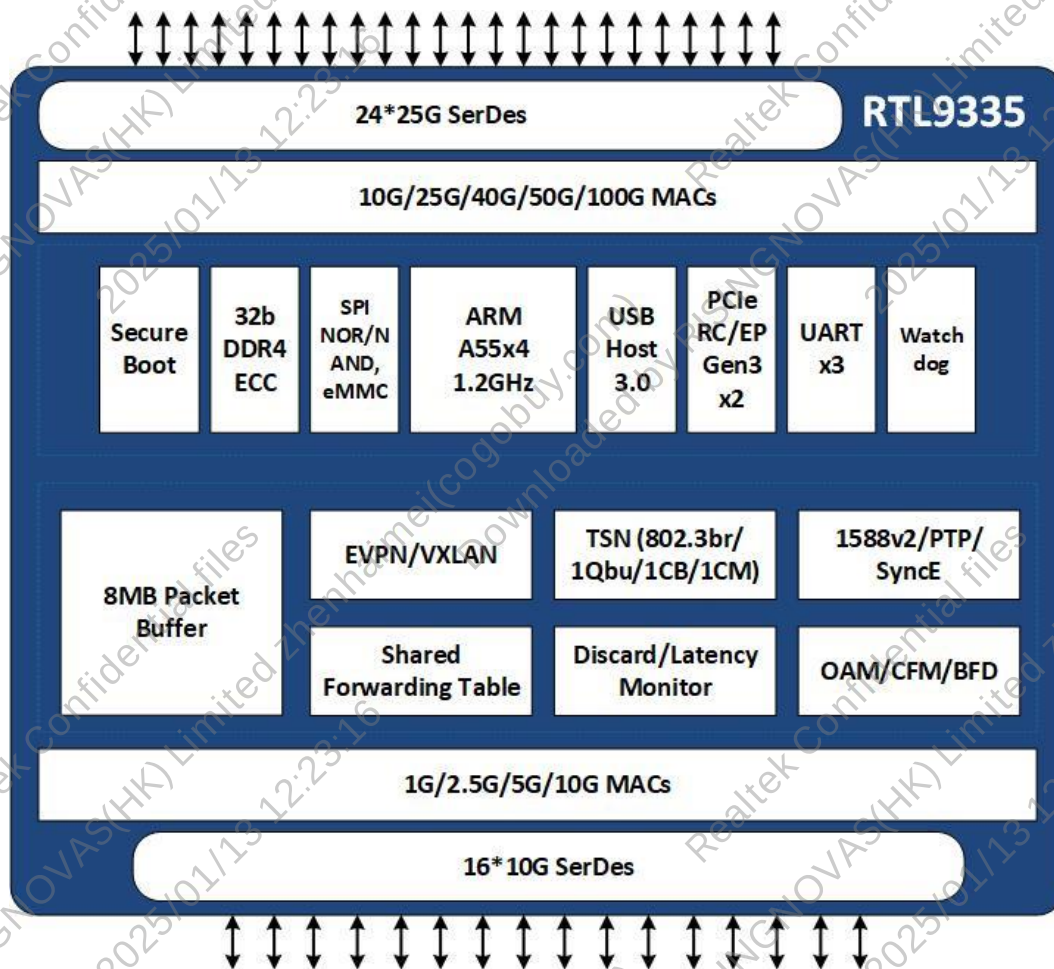


Figure 1. RTL9335 Block Diagram

4. Pin Assignments

4.1. RTL9335 Pin Assignments (Top View)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
A	S0ND	S0ND	HS0N_00	HS0P_00	S0ND	S0ND	S0ND	HSIN_00	HSIP_00	S0ND	SVDD_CK0	D0ND	RESERVED_A1	SYNC_OUT_C LK_BKUP	D0ND	SYNC_OUT_C LK	D0ND	MDC_0	MDC_1
B	HS0N_01	HS0P_01	S0ND	S0ND	SVDD_TRX	HSIN_01	HSIP_01	S0ND	S0ND	SVDD_CK1	S0ND	REXT(ORREF)	RESERVED_B1	SYNC_OUT_C OCK_BRUP	SCL3	SYNC_OUT_C OCK	LED_DATA/ LED_M0NO	MDIO_0	MDIO_1
C	S0ND	S0ND	HS0N_02	HS0P_02	S0ND	S0ND	S0ND	HSIN_02	HSIP_02	S0ND	SVDD_CK2	D0ND	RESERVED_C1	RESETB	SDA3	D0ND	SCL2	D0ND	GPIO_MDC_A
D	HS0N_03	HS0P_03	S0ND	S0ND	SVDD_TRX	HSIN_03	HSIP_03	S0ND	S0ND	SVDD_CK3	S0ND	RESERVED_D1	RESERVED_D1	PWRMON	HW_RESET_5	SDA2	RESERVED_D1	LED_CLK/ LED_MDC	GPIO_MDIO_A
E	S0ND	S0ND	HS0N_04	HS0P_04	S0ND	S0ND	S0ND	HSIN_04	HSIP_04	S0ND	SVDD_CK4	D0ND	RESERVED_E1	D0ND	RESERVED_E1	SOC_PCIE_MO DE_SEL[1]	RESERVED_E1	GPIO[11]/ LED_SYNC	D0ND
F	HS0N_05	HS0P_05	S0ND	S0ND	SVDD_TRX	HSIN_05	HSIP_05	S0ND	S0ND	SVDD_CK5	S0ND	RESERVED_F1	RESERVED_F1	RESERVED_F1	RST_OUT	RESERVED_F1	RESERVED_F1	GPIO[19]/ LED_STACK	PCIE_RSTB
G	S0ND	S0ND	HS0N_06	HS0P_06	S0ND	S0ND	S0ND	HSIN_06	HSIP_06	S0ND	SVDD_CK6	D0ND	RESERVED_G1	RESERVED_G1	RESERVED_G1	RESERVED_G1	SOC_PCIE_MO DE_SEL[0]	RESERVED_G1	GPIO[0]/ SYS_LED
H	HS0N_07	HS0P_07	S0ND	S0ND	SVDD_TRX	HSIN_07	HSIP_07	S0ND	S0ND	SVDD_CK7	S0ND	AVDDH (B0)	AVDDV10 (B0)	AVDDV09 (B0)	D0ND	D0ND	D0ND	D0ND	D0ND
I	S0ND	S0ND	HS0N_08	HS0P_08	S0ND	S0ND	S0ND	HSIN_08	HSIP_08	S0ND	SVDD_CK8	A0ND	A0ND	A0ND	D0ND	D0ND	D0ND	D0ND	D0ND
K	HS0N_09	HS0P_09	S0ND	S0ND	SVDD_TRX	HSIN_09	HSIP_09	S0ND	S0ND	SVDD_CK9	S0ND	D0ND	D0ND	D0ND	D0ND	D0ND	DVDDH	DVDDH	DVDDH
L	S0ND	S0ND	HS0N_10	HS0P_10	S0ND	S0ND	S0ND	HSIN_10	HSIP_10	S0ND	SVDD_CK10	D0ND	D0ND	D0ND	D0ND	D0ND	D0ND	D0ND	D0ND
M	HS0N_11	HS0P_11	S0ND	S0ND	SVDD_TRX	HSIN_11	HSIP_11	S0ND	S0ND	SVDD_CK11	S0ND	D0ND	DVDDL	D0ND	DVDDL	D0ND	DVDDL	D0ND	DVDDL
N	S0ND	S0ND	HS0N_12	HS0P_12	S0ND	S0ND	S0ND	HSIN_12	HSIP_12	S0ND	SVDD_CK12	D0ND	DVDDL	D0ND	DVDDL	D0ND	DVDDL	D0ND	DVDDL
P	HS0N_13	HS0P_13	S0ND	S0ND	SVDD_TRX	HSIN_13	HSIP_13	S0ND	S0ND	SVDD_CK13	S0ND	D0ND	DVDDL	D0ND	DVDDL	D0ND	DVDDL	D0ND	DVDDL
R	S0ND	S0ND	HS0N_14	HS0P_14	S0ND	S0ND	S0ND	HSIN_14	HSIP_14	S0ND	SVDD_CK14	D0ND	DVDDL	D0ND	DVDDL	D0ND	DVDDL	D0ND	DVDDL
T	HS0N_15	HS0P_15	S0ND	S0ND	SVDD_TRX	HSIN_15	HSIP_15	S0ND	S0ND	SVDD_CK15	S0ND	D0ND	DVDDL	D0ND	DVDDL	D0ND	DVDDL	D0ND	DVDDL
U	S0ND	S0ND	S0ND	S0ND	S0ND	S0ND	S0ND	S0ND	S0ND	S0ND	S0ND	D0ND	DVDDL	D0ND	DVDDL	D0ND	DVDDL	D0ND	DVDDL
V	HS0P_16	HS0N_16	S0ND	HSIP_16	HSIN_16	S0ND	S0ND	S0ND	S0ND	D0ND	DVDDL	D0ND	DVDDL	D0ND	DVDDL	D0ND	DVDDL	D0ND	DVDDL
W	S0ND	S0ND	S0ND	S0ND	S0ND	S0ND	S0ND	S0ND	S0ND	D0ND	DVDDL	D0ND	DVDDL	D0ND	DVDDL	D0ND	DVDDL	D0ND	DVDDL

Figure 2. RTL9335 Pin Assignments (Top View)



20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	
DGND	SPI_SDO/SIO1	NAND_10[6]/SD_DATA6	NAND_10[0]/SD_DATA0	NAND_WE_N	NAND_CE[1]/SD_CMD	AGND	USB_HSP0	AGND	USB_HSP1	AGND	SOMIL_HSP0	DGND	DM0#DBIL#	DQ17	DQ19	DQ83B	DQ83	NA	A
MDC_OOB	SPI_SIO2	SPI_CS[1]	NAND_10[1]/SD_DATA1	DGND	NAND_CE[0]/SD_CLK	AGND	USB_HSON	AGND	USB_HSIN	AGND	SOMIL_HSON	DGND	DGND	DQ21	DQ23	DGND	DQ25	DQ27	B
MDIO_OOB	SPI_SIO3	DGND	NAND_10[2]/SD_DATA2	NAND_10[4]/SD_DATA4	NAND_RE_N	NAND_RB_N/SD_DS	AGND	USB_VDDL	AGND	SVDDL_OOB_TRX	AGND	DGND	DQ18	DGND	DQ22	DQ24	DQ31	NA	C
DGND	SPI_SCLK	SPI_CS[0]	NAND_10[7]/SD_DATA7	NAND_10[5]/SD_DATA5	NAND_CLE	AGND	PCIE_CLKP	AGND	USB_HSDM	AGND	SOMIL_HSP1	DGND	DGND	DQ16	DQ20	DM3#DBIU#	DQ28	DQ30	D
GPIO[16]/UART2_TXD	SPI_RSTN	SPI_SDI/SIO0	DGND	NAND_10[3]/SD_DATA3	NAND_RE_N/SP_RSTN	AGND	PCIE_CLKN	AGND	USB_HSPF	AGND	SOMIL_HSIN	DGND	TP_E33	DQ26	DQ32	DQ34	DQ36	NA	E
RESERVED_F20	GPIO[17]/UART2_RXD	GPIO[15]/PTT_ONS_PFS_IN	GPIO[13]/PTT_CLOCK_OUTPUT	NAND_WE_N	DGND	PCIE_CK	AGND	USB_VDDH	AGND	PCIE_VDDL	AGND	DGND	DGND	DGND	DGND	DGND	DGND	DGND	F
PHY_SYS_STO_P_PCIECK	DDR_BANKSEL	GPIO[16]/PTT_TOD_OUT	GPIO[14]/PTT_ONS_PFS_OUT	DGND	GPIO[12]/PTT_TIME_OF	AGND	PCIE_HSP0	AGND	PCIE_HSP1	AGND	PCIE_HSP1	DGND	DM0#DBIL#	DQ1	DQ3	DQ81B	DQ81	NA	G
DGND	DGND	DGND	DGND	DGND	DVDD_F	AGND	PCIE_HSON0	AGND	PCIE_HSIN1	AGND	PCIE_HSON1	DGND	DGND	DQ5	DQ7	DGND	DQ9	DQ11	H
DGND	DVDD_MDXX_0	DVDD_MDXX_1	DVDD_MDXX_OOB	DGND	DVDD_F	DVDD_IO	AGND	PCIE_HSP0	AGND	PCIE_VDDP	AGND	DGND	DQ2	DGND	DQ6	DQ13	DQ15	NA	I
DVDDH	DGND	DVDD_NF	DVDD_NF	DGND	DGND	DVDD_IO	AGND	PCIE_HSON0	AGND	AGND	AGND	DGND	DGND	DQ0	DQ4	DM1#DBIU#	DQ12	DQ14	K
DGND	DGND	DGND	DGND	DGND	DGND	DGND	DGND	AGND	AGND	AVDD09_PLL_CPU	AVDD33_PLL_CPU	DGND	TP_L33	DQ30B	DQ30	DQ8	DQ10	NA	L
DGND	DVDDL	DGND	DVDDL	DGND	DVDDL	DGND	DVDDL	DGND	DVDDH	TP_M30	TP_M31	DGND	DGND	DGND	DGND	DGND	DGND	DGND	M
DGND	DVDDL	DGND	DVDDL	DGND	DVDDL	DGND	DVDDL_CPU	DVDDL_CPU	DGND	DGND	DGND	DGND	DGND	ECC1	ECC3	ECC_DQ80B	ECC_DQ80	NA	N
DGND	DVDDL	DGND	DVDDL	DGND	DVDDL	DVDDL_CPU	DVDDL_CPU	DVDDL_CPU	DGND	MVDD	MVDDH	DGND	TP_F39	DGND	DGND	DGND	ECC0	ECC2	P
DGND	DVDDL	DGND	DVDDL	DGND	DVDDL	DGND	DVDDL_CPU	DGND	DGND	MVDD	DGND	DGND	ALERT1	RSTN	CSN	TP_R36	TP_R37	NA	R
DGND	DVDDL	DGND	DVDDL	DGND	DVDDL	DGND	DVDDL	DGND	MVDD	DGND	MVDD	DGND	DGND	AS0#A16	BG1	DGND	CLK1B	CLK1	T
DGND	DVDDL	DGND	DVDDL	DGND	DVDDL	DGND	DVDDL	DGND	DGND	MVDD	DGND	DGND	DGND	CKE1	GA0#A15	BA1	CLK8	CLK	U
DGND	DVDDL	DGND	DVDDL	DGND	DVDDL	DGND	DVDDL	DGND	MVDD_CK	DGND	MVDD	DGND	CSIN	TEN	DGND	A9	CKE	NA	V
DGND	DVDDL	DGND	DVDDL	DGND	DVDDL	DGND	DVDDL	DGND	DGND	MVDD	DGND	DGND	DGND	A5	ODT0	A1	A7	W	

Figure 3. RTL9335 Pin Assignments (Top View) (Continued)



D0ND	DVDDL	D0ND	DVDDL	D0ND	DVDDL	D0ND	DVDDL	D0ND	MVDD	D0ND	MVDD	D0ND	WE#A14	ACTN	D0ND	A0	A12/BC#	NA	Y
D0ND	DVDDL	D0ND	DVDDL	D0ND	DVDDL	D0ND	DVDDL	D0ND	D0ND	MVDD	D0ND	D0ND	D0ND	D0ND	A13	A11	A2	A10/AF	AA
D0ND	DVDDL	D0ND	DVDDL	D0ND	DVDDL	D0ND	DVDDL	D0ND	MVDD	D0ND	MVDD	D0ND	ALERT0	NA	D0ND	B00	A3	NA	AB
D0ND	DVDDL	D0ND	DVDDL	D0ND	DVDDL	D0ND	DVDDL	D0ND	D0ND	D0ND	D0ND	D0ND	ODT1	ALERT2	A8	BA0	PAR	A6	AC
D0ND	DVDDL	D0ND	DVDDL	D0ND	DVDDL	D0ND	DVDDL	DVDDH	D0ND	D0ND	D0ND	D0ND	TP_AD32	TP_AD33	GPIO[31]	DRAM_SEL	D0ND	D0ND	AD
D0ND	DVDDL	D0ND	DVDDL	D0ND	DVDDL	D0ND	DVDDL	DVDDH	D0ND	DVDD_GPIO	D0ND	D0ND	GPIO[26]	GPIO[27]	GPIO[28]	GPIO[29]	GPIO[30]	RESERVED_A8_38	AE
D0ND	DVDDL	D0ND	DVDDL	D0ND	DVDDL	D0ND	DVDDL	D0ND	D0ND	D0ND	DVDD_ITA0	D0ND	GPIO[23]	GPIO[24]	GPIO[25]	D0R_DATA_B_US_SEL	RESERVED_AF_37	RESERVED_AF_38	AF
S0ND	S0ND	S0ND	S0ND	S0ND	S0ND	S0ND	S0ND	D0ND	D0ND	DVDD_SF1	D0ND	D0ND	GPIO[21]	GPIO[22]	D0ND	GPIO[6]/JTAG_TDO	SYS_LED_EN	DDR_ECC_BU_S_EN	AG
S0ND	S0ND	S0ND	S0ND	S0ND	S0ND	S0ND	S0ND	D0ND	D0ND	DVDD_MDIO_3	D0ND	GPIO[111]/MSPL_CS#1	GPIO[20]	GPIO[2]/JTAG_TMS	GPIO[3]/JTAG_TCK	GPIO[4]/JTAG_TRST#	GPIO[5]/JTAG_TDI	AH	
S0ND	SVDDH_TX_25G	S0ND	S0ND	S0ND	SVDDH_25G	S0ND	S0ND	S0ND	D0ND	DVDD_MDIO_2	AGND	XTAL_AV09	D0ND	GPIO[8]/MSPL_MISO	GPIO[9]/MSPL_MOSI	GPIO[10]/MSPL_CS#0	GPIO_MDC_B	GPIO_MDIO_B	AI
SVDDTRX_25G	S0ND	SVDDTRX_25G	S0ND	SVDDTRX_25G	S0ND	SVDDTRX_25G	S0ND	SVDDTRX_25G	D0ND	D0ND	AGND	XTAL_AV33	D0ND	GPIO[7]/MSPL_SCK	UART0_TXD	TP_AK36	D0ND	TP_AK38	AK
S0ND	S0ND	S0ND	S0ND	S0ND	S0ND	S0ND	S0ND	S0ND	D0ND	AVDD09_FLL	AGND	AGND	USB_LED	UART0_RXD	D0ND	TP_AL36	TP_AL37	TP_AL38	AL
SVDDCK_LQ_07_25G	S0ND	SVDDCK_HQ_08_25G	S0ND	SVDDCK_LQ_08_25G	S0ND	SVDDCK_HQ_09_25G	S0ND	SVDDCK_LQ_09_25G	D0ND	AVDD33_FLL	AGND	AVDD33_25G_B0	RESERVED_A_M33	UART0_CTS#	UART0_RTS#	SCL1	SDA1	D0ND	AM
S0ND	S0ND	S0ND	S0ND	S0ND	S0ND	S0ND	S0ND	S0ND	S0ND	S0ND	S0ND	S0ND	PTP_CLK_IN	RESERVED_A_N34	UART1_TXD	D0ND	SDA0	SCL0	AN
S0ND	HSIN_S34	S0ND	HSIN_S35	S0ND	HSIN_S36	S0ND	HSIN_S37	S0ND	HSIN_S38	S0ND	HSIN_S39	S0ND	PTP_CLKN	UART1_CTS#	UART1_RXD	MDIO_3	MDC_2	MDIO_2	AP
S0ND	HSIF_S34	S0ND	HSIF_S35	S0ND	HSIF_S36	S0ND	HSIF_S37	S0ND	HSIF_S38	S0ND	HSIF_S39	S0ND	PTP_CLKP	D0ND	UART1_RTS#	MDC_3	D0ND	RESERVED_A_R38	AR
S0ND	S0ND	S0ND	S0ND	S0ND	S0ND	S0ND	S0ND	S0ND	S0ND	S0ND	S0ND	S0ND	D0ND	PTP_CLK_S	25G_REXT (BREF)	RESERVED_AT_36	TP_AT37	TP_AT38	AT
HSON_S33	S0ND	HSON_S34	S0ND	HSON_S35	S0ND	HSON_S36	S0ND	HSON_S37	S0ND	HSON_S38	S0ND	HSON_S39	S0ND	REFCLKN_156P125	D0ND	BOOT_SEL[1]	BOOT_SEL[0]	RESERVED_A_U38	AU
HROP_S33	S0ND	HROP_S34	S0ND	HROP_S35	S0ND	HROP_S36	S0ND	HROP_S37	S0ND	HROP_S38	S0ND	HROP_S39	S0ND	REFCLKP_156P125	D0ND	X0	X1	D0ND	AV
20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	

Figure 4. RTL9335 Pin Assignments (Top View) (Continued)

Y	HSOP_S17	HSON_S17	SGND	HSIF_S17	HSIN_S17	SGND	SGND	SGND	SGND	DGND	DVDDL	DGND	DVDDL	DGND	DVDDL	DGND	DVDDL	DGND	DVDDL
AA	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	DGND	DVDDL	DGND	DVDDL	DGND	DVDDL	DGND	DVDDL	DGND	DVDDL
AB	HSOP_S18	HSON_S18	SGND	HSIF_S18	HSIN_S18	SGND	SGND	SGND	SGND	DGND	DVDDL	DGND	DVDDL	DGND	DVDDL	DGND	DVDDL	DGND	DVDDL
AC	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	DGND	DVDDL	DGND	DVDDL	DGND	DVDDL	DGND	DVDDL	DGND	DVDDL
AD	HSOP_S19	HSON_S19	SGND	HSIF_S19	HSIN_S19	SGND	SGND	SGND	SGND	DGND	DVDDL	DGND	DVDDL	DGND	DVDDL	DGND	DVDDL	DGND	DVDDL
AE	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	DGND	DVDDL	DGND	DVDDL	DGND	DVDDL	DGND	DVDDL	DGND	DVDDL
AF	HSOP_S20	HSON_S20	SGND	HSIF_S20	HSIN_S20	SGND	SGND	SGND	SGND	DGND	DVDDL	DGND	DVDDL	DGND	DVDDL	DGND	DVDDL	DGND	DVDDL
AG	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND
AH	HSOP_S21	HSON_S21	SGND	HSIF_S21	HSIN_S21	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND
AI	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SVDDH_RX_25G	SGND	SVDDH_RX_25G	SGND	SGND
AK	HSOP_S22	HSON_S22	SGND	HSIF_S22	HSIN_S22	SGND	SGND	SGND	SGND	SVDDCK_HQ_04_25G	SGND	SVDDCK_LQ_04_25G	SGND	SVDDTRX_25G	SGND	SVDDTRX_25G	SGND	SVDDTRX_25G	SGND
AL	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND
AM	HSOP_S23	HSON_S23	SGND	HSIF_S23	HSIN_S23	SGND	SGND	SGND	SGND	SVDDCK_HQ_05_25G	SGND	SVDDCK_LQ_05_25G	SGND	SVDDCK_HQ_06_25G	SGND	SVDDCK_LQ_06_25G	SGND	SVDDCK_HQ_07_25G	SGND
AN	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND
AP	HSIN_S24	SGND	HSIN_S25	SGND	HSIN_S26	SGND	HSIN_S27	SGND	HSIN_S28	SGND	HSIN_S29	SGND	HSIN_S30	SGND	HSIN_S31	SGND	HSIN_S32	SGND	HSIN_S33
AR	HSIF_S24	SGND	HSIF_S25	SGND	HSIF_S26	SGND	HSIF_S27	SGND	HSIF_S28	SGND	HSIF_S29	SGND	HSIF_S30	SGND	HSIF_S31	SGND	HSIF_S32	SGND	HSIF_S33
AT	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND	SGND
AU	SGND	HSON_S24	SGND	HSON_S25	SGND	HSON_S26	SGND	HSON_S27	SGND	HSON_S28	SGND	HSON_S29	SGND	HSON_S30	SGND	HSON_S31	SGND	HSON_S32	SGND
AV	SGND	HSOP_S24	SGND	HSOP_S25	SGND	HSOP_S26	SGND	HSOP_S27	SGND	HSOP_S28	SGND	HSOP_S29	SGND	HSOP_S30	SGND	HSOP_S31	SGND	HSOP_S32	SGND
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19

Figure 5. RTL9335 Pin Assignments (Top View) (Continued)

4.2. Pin Assignments Table Definitions

I: Input Pin

O: Output Pin

I/O: Bi-Direction Input/Output Pin

DP: Digital Power Pin

DG: Digital Ground Pin

SP: SERDES Power Pin

MP: DDR Power Pin

AI: Analog Input Pin

AO: Analog Output Pin

AI/O: Analog Bi-Direction Input/Output Pin

AP: Analog Power Pin

AG: Analog Ground Pin

SG: SERDES Ground Pin

4.3. RTL9335 Pin Assignments Table

Table 1. RTL9335 Pin Assignments Table

Pin Name	Pin No.	Type
SGND	A1	SG
SGND	A2	SG
HS0N_S0	A3	AO
HS0P_S0	A4	AO
SGND	A5	SG
SGND	A6	SG
SGND	A7	SG
HSIN_S0	A8	AI
HSIP_S0	A9	AI
SGND	A10	SG
SVDD_CK0	A11	SP
DGND	A12	DG
RESERVED_A13	A13	
SYNC_OUT_CLK_BKUP	A14	O
DGND	A15	DG
SYNC_OUT_CLK	A16	O
DGND	A17	DG
MDC_0	A18	O
MDC_1	A19	O
DGND	A20	DG
SPI_SDO/SIO1	A21	I/O
NAND_IO[6] / SD_DAT6	A22	I/O

Pin Name	Pin No.	Type
NAND_IO[0] / SD_DAT0	A23	I/O
NAND_WE_N	A24	O
NAND_CE#[1] / SD_CMD	A25	I/O
AGND	A26	AG
USB_HSOP	A27	AO
AGND	A28	AG
USB_HSIP	A29	AI
AGND	A30	AG
SGMII_HSOP	A31	AO
DGND	A32	DG
DM2#/DBIL#	A33	O
DQ17	A34	I/O
DQ19	A35	I/O
DQS3B	A36	I/O
DQS3	A37	I/O
	A38	
HS0N_S1	B1	AO
HS0P_S1	B2	AO
SGND	B3	SG
SGND	B4	SG
SVDD_TRX	B5	SP
HSIN_S1	B6	AI

Pin Name	Pin No.	Type
HSIP_S1	B7	AI
SGND	B8	SG
SGND	B9	SG
SVDD_CK1	B10	SP
SGND	B11	SG
REXT (IBREF)	B12	AI/O
RESERVED_B13	B13	
SYNC_OUT_LOCK_BKUP	B14	O
SCL3	B15	O
SYNC_OUT_LOCK	B16	O
LED_DATA / LED_MDIO	B17	I/O
MDIO_0	B18	I/O
MDIO_1	B19	I/O
MDC_OOB	B20	O
SPI_SIO2	B21	I/O
SPI_CS#[1]	B22	O
NAND_IO[1] / SD_DAT1	B23	I/O
DGND	B24	DG
NAND_CE#[0] / SD_CLK	B25	O
AGND	B26	AG
USB_HSON	B27	AO
AGND	B28	AG
USB_HSIN	B29	AI
AGND	B30	AG
SGMII_HSON	B31	AO
DGND	B32	DG
DGND	B33	DG
DQ21	B34	I/O
DQ23	B35	I/O
DGND	B36	DG
DQ25	B37	I/O
DQ27	B38	I/O
SGND	C1	SG
SGND	C2	SG
HSOP_S2	C3	AO
HSOP_S2	C4	AO
SGND	C5	SG
SGND	C6	SG
SGND	C7	SG
HSIN_S2	C8	AI
HSIP_S2	C9	AI
SGND	C10	SG
SVDD_CK2	C11	SP
DGND	C12	DG

Pin Name	Pin No.	Type
RESERVED_C13	C13	
RESETB	C14	AI
SDA3	C15	IO
DGND	C16	DG
SCL2	C17	O
DGND	C18	DG
GPIO_MDC_A	C19	O
MDIO_OOB	C20	I/O
SPI_SIO3	C21	I/O
DGND	C22	DG
NAND_IO[2] / SD_DAT2	C23	I/O
NAND_IO[4] / SD_DAT4	C24	I/O
NAND_RE_N	C25	O
NAND_RB_N / SD_DS	C26	I
AGND	C27	AG
USB_VDDL	C28	AP
AGND	C29	AG
SVDDL_OOB_TRX	C30	SP
AGND	C31	AG
DGND	C32	DG
DQ18	C33	I/O
DGND	C34	DG
DQ22	C35	I/O
DQ29	C36	I/O
DQ31	C37	I/O
	C38	
HSOP_S3	D1	AO
HSOP_S3	D2	AO
SGND	D3	SG
SGND	D4	SG
SVDD_TRX	D5	SP
HSIN_S3	D6	AI
HSIP_S3	D7	AI
SGND	D8	SG
SGND	D9	SG
SVDD_CK3	D10	SP
SGND	D11	SG
RESERVED_D12	D12	
RESERVED_D13	D13	
PWRMON	D14	AI
HW_RESET_EN	D15	I
SDA2	D16	I/O
RESERVED_D17	D17	
LED_CLK / LED_MDC	D18	O

Pin Name	Pin No.	Type
GPIO_MDIO_A	D19	I/O
DGND	D20	DG
SPI_SCLK	D21	O
SPI_CS#[0]	D22	O
NAND_IO[7] / SD_DAT7	D23	I/O
NAND_IO[5] / SD_DAT5	D24	I/O
NAND_CLE	D25	O
AGND	D26	AG
PCIE_CLKP	D27	AI/O
AGND	D28	AG
USB_HSDM	D29	AI/O
AGND	D30	AG
SGMII_HSIP	D31	AI
DGND	D32	DG
DGND	D33	DG
DQ16	D34	I/O
DQ20	D35	I/O
DM3#/DBIU#	D36	O
DQ28	D37	I/O
DQ30	D38	I/O
SGND	E1	SG
SGND	E2	SG
HSO_N_S4	E3	AO
HSO_P_S4	E4	AO
SGND	E5	SG
SGND	E6	SG
SGND	E7	SG
HSI_N_S4	E8	AI
HSI_P_S4	E9	AI
SGND	E10	SG
SVDD_CK4	E11	SP
DGND	E12	DG
RESERVED_E13	E13	
DGND	E14	DG
RESERVED_E15	E15	
SOC_PCIE_MODE_SEL[1]	E16	I
RESERVED_E17	E17	
GPIO [1] / LED_SYNC	E18	I/O
DGND	E19	DG
GPIO [18] / UART2_TXD	E20	I/O
SPI_RSTN	E21	O
SPI_SDI/SIO0	E22	I/O
DGND	E23	DG
NAND_IO[3] / SD_DAT3	E24	I/O

Pin Name	Pin No.	Type
NAND_ALE / SD_RST_N	E25	O
AGND	E26	AG
PCIE_CLKN	E27	AI/O
AGND	E28	AG
USB_HSDP	E29	AI/O
AGND	E30	AG
SGMII_HSI_N	E31	AI
DGND	E32	DG
TP_E33	E33	
DQS2B	E34	I/O
DQS2	E35	I/O
DQ24	E36	I/O
DQ26	E37	I/O
	E38	
HSO_N_S5	F1	AO
HSO_P_S5	F2	AO
SGND	F3	SG
SGND	F4	SG
SVDD_TRX	F5	SP
HSI_N_S5	F6	AI
HSI_P_S5	F7	AI
SGND	F8	SG
SGND	F9	SG
SVDD_CK5	F10	SP
SGND	F11	SG
RESERVED_F12	F12	
RESERVED_F13	F13	
RESERVED_F14	F14	
RST_OUT	F15	O
RESERVED_F16	F16	
RESERVED_F17	F17	
GPIO [19] / LED_STACK	F18	I/O
PCIE_RSTB	F19	I/O
RESERVED_F20	F20	
GPIO [17] / UART2_RXD	F21	I/O
GPIO [15] / PTP_ONE_PPS_IN	F22	I/O
GPIO [13] / PTP_CLOCK_OUTPUT	F23	I/O
NAND_WP_N	F24	O
DGND	F25	DG
PCIE_CK	F26	AP
AGND	F27	AG
USB_VDDH	F28	AP
AGND	F29	AG
PCIE_VDDL	F30	AP

Pin Name	Pin No.	Type
AGND	F31	AG
DGND	F32	DG
DGND	F33	DG
DGND	F34	DG
DGND	F35	DG
DGND	F36	DG
DGND	F37	DG
DGND	F38	DG
SGND	G1	SG
SGND	G2	SG
HSO _N _S6	G3	AO
HSO _P _S6	G4	AO
SGND	G5	SG
SGND	G6	SG
SGND	G7	SG
HSIN _{_S6}	G8	AI
HSIP _{_S6}	G9	AI
SGND	G10	SG
SVDD _{_CK6}	G11	SP
DGND	G12	DG
RESERVED _{_G13}	G13	
RESERVED _{_G14}	G14	
RESERVED _{_G15}	G15	
RESERVED _{_G16}	G16	
SOC_PCIE_MODE_SEL[0]	G17	I
RESERVED _{_G18}	G18	
GPIO [0] / SYS_LED	G19	I/O
PHY_SYS_STOP_PCIECK	G20	I
DDR_RANK_SEL	G21	I
GPIO [16] / PTP_TOD_OUTPUT	G22	I/O
GPIO [14] / PTP_ONE_PPS_OUT	G23	I/O
DGND	G24	DG
GPIO [12] / PTP_TIME_OP	G25	I/O
AGND	G26	AG
PCIE_HSO _{P0}	G27	AO
AGND	G28	AG
PCIE_HSI _{P1}	G29	AI
AGND	G30	AG
PCIE_HSO _{P1}	G31	AO
DGND	G32	DG
DM0#/DBIL#	G33	O
DQ1	G34	I/O
DQ3	G35	I/O
DQS1B	G36	I/O

Pin Name	Pin No.	Type
DQS1	G37	I/O
	G38	
HSO _N _S7	H1	AO
HSO _P _S7	H2	AO
SGND	H3	SG
SGND	H4	SG
SVDD _{_TRX}	H5	SP
HSIN _{_S7}	H6	AI
HSIP _{_S7}	H7	AI
SGND	H8	SG
SGND	H9	SG
SVDD _{_CK7}	H10	SP
SGND	H11	SG
AVDDH (BG)	H12	AP
AVDDV10 (BG)	H13	AP
AVDDV09 (BG)	H14	AP
DGND	H15	DG
DGND	H16	DG
DGND	H17	DG
DGND	H18	DG
DGND	H19	DG
DGND	H20	DG
DGND	H21	DG
DGND	H22	DG
DGND	H23	DG
DGND	H24	DG
DVDD _{_P}	H25	DP
AGND	H26	AG
PCIE_HSO _{N0}	H27	AO
AGND	H28	AG
PCIE_HSI _{N1}	H29	AI
AGND	H30	AG
PCIE_HSO _{N1}	H31	AO
DGND	H32	DG
DGND	H33	DG
DQ5	H34	I/O
DQ7	H35	I/O
DGND	H36	DG
DQ9	H37	I/O
DQ11	H38	I/O
SGND	J1	SG
SGND	J2	SG
HSO _N _S8	J3	AO
HSO _P _S8	J4	AO

Pin Name	Pin No.	Type
SGND	J5	SG
SGND	J6	SG
SGND	J7	SG
HSIN_S8	J8	AI
HSIP_S8	J9	AI
SGND	J10	SG
SVDD CK8	J11	SP
AGND	J12	AG
AGND	J13	AG
AGND	J14	AG
DGND	J15	DG
DGND	J16	DG
DGND	J17	DG
DGND	J18	DG
DGND	J19	DG
DGND	J20	DG
DVDD_MDX_0	J21	DP
DVDD_MDX_1	J22	DP
DVDD_MDX_OOB	J23	DP
DGND	J24	DG
DVDD_P	J25	DP
DVDD_IO	J26	DP
AGND	J27	AG
PCIE_HSIPO	J28	AI
AGND	J29	AG
PCIE_VDDL	J30	AP
AGND	J31	AG
DGND	J32	DG
DQ2	J33	I/O
DGND	J34	DG
DQ6	J35	I/O
DQ13	J36	I/O
DQ15	J37	I/O
	J38	
HSIN_S9	K1	AO
HSOP_S9	K2	AO
SGND	K3	SG
SGND	K4	SG
SVDD_TRX	K5	SP
HSIN_S9	K6	AI
HSIP_S9	K7	AI
SGND	K8	SG
SGND	K9	SG
SVDD CK9	K10	SP

Pin Name	Pin No.	Type
SGND	K11	SG
DGND	K12	DG
DGND	K13	DG
DGND	K14	DG
DGND	K15	DG
DGND	K16	DG
DVDDH	K17	DP
DVDDH	K18	DP
DVDDH	K19	DP
DVDDH	K20	DP
DGND	K21	DG
DVDD_NF	K22	DP
DVDD_NF	K23	DP
DGND	K24	DG
DGND	K25	DG
DVDD_IO	K26	DP
AGND	K27	AG
PCIE_HSIN0	K28	AI
AGND	K29	AG
AGND	K30	AG
AGND	K31	AG
DGND	K32	DG
DGND	K33	DG
DQ0	K34	I/O
DQ4	K35	I/O
DM1#/DBIU#	K36	O
DQ12	K37	I/O
DQ14	K38	I/O
SGND	L1	SG
SGND	L2	SG
HSIN_S10	L3	AO
HSOP_S10	L4	AO
SGND	L5	SG
SGND	L6	SG
SGND	L7	SG
HSIN_S10	L8	AI
HSIP_S10	L9	AI
SGND	L10	SG
SVDD CK10	L11	SP
DGND	L12	DG
DGND	L13	DG
DGND	L14	DG
DGND	L15	DG
DGND	L16	DG

Pin Name	Pin No.	Type
DGND	L17	DG
DGND	L18	DG
DGND	L19	DG
DGND	L20	DG
DGND	L21	DG
DGND	L22	DG
DGND	L23	DG
DGND	L24	DG
DGND	L25	DG
DGND	L26	DG
DGND	L27	DG
AGND	L28	AG
AGND	L29	AG
AVDD09_PLL_CPU	L30	AP
AVDD33_PLL_CPU	L31	AP
DGND	L32	DG
TP_L33	L33	
DQS0B	L34	I/O
DQS0	L35	I/O
DQ8	L36	I/O
DQ10	L37	I/O
	L38	
HS0N_S11	M1	AO
HS0P_S11	M2	AO
SGND	M3	SG
SGND	M4	SG
SVDD_TRX	M5	SP
HSIN_S11	M6	AI
HSIP_S11	M7	AI
SGND	M8	SG
SGND	M9	SG
SVDD_CK11	M10	SP
SGND	M11	SG
DGND	M12	DG
DVDDL	M13	DP
DGND	M14	DG
DVDDL	M15	DP
DGND	M16	DG
DVDDL	M17	DP
DGND	M18	DG
DVDDL	M19	DP
DGND	M20	DG
DVDDL	M21	DP
DGND	M22	DG

Pin Name	Pin No.	Type
DVDDL	M23	DP
DGND	M24	DG
DVDDL	M25	DP
DGND	M26	DG
DVDDL	M27	DP
DGND	M28	DG
DVDDH	M29	DP
TP_M30	M30	
TP_M31	M31	
DGND	M32	DG
DGND	M33	DG
DGND	M34	DG
DGND	M35	DG
DGND	M36	DG
DGND	M37	DG
DGND	M38	DG
SGND	N1	SG
SGND	N2	SG
HS0N_S12	N3	AO
HS0P_S12	N4	AO
SGND	N5	SG
SGND	N6	SG
SGND	N7	SG
HSIN_S12	N8	AI
HSIP_S12	N9	AI
SGND	N10	SG
SVDD_CK12	N11	SP
DGND	N12	DG
DVDDL	N13	DP
DGND	N14	DG
DVDDL	N15	DP
DGND	N16	DG
DVDDL	N17	DP
DGND	N18	DG
DVDDL	N19	DP
DGND	N20	DG
DVDDL	N21	DP
DGND	N22	DG
DVDDL	N23	DP
DGND	N24	DG
DVDDL	N25	DP
DGND	N26	DG
DVDDL_CPU	N27	DP
DVDDL_CPU	N28	DP

Pin Name	Pin No.	Type
DGND	N29	DG
DGND	N30	DG
DGND	N31	DG
DGND	N32	DG
DGND	N33	DG
ECC1	N34	I/O
ECC3	N35	I/O
ECC_DQS0B	N36	I/O
ECC_DQS0	N37	I/O
	N38	
HSO_N_S13	P1	AO
HSO_P_S13	P2	AO
SGND	P3	SG
SGND	P4	SG
SVDD_TRX	P5	SP
HSIN_S13	P6	AI
HSIP_S13	P7	AI
SGND	P8	SG
SGND	P9	SG
SVDD_CK13	P10	SP
SGND	P11	SG
DGND	P12	DG
DVDDL	P13	DP
DGND	P14	DG
DVDDL	P15	DP
DGND	P16	DG
DVDDL	P17	DP
DGND	P18	DG
DVDDL	P19	DP
DGND	P20	DG
DVDDL	P21	DP
DGND	P22	DG
DVDDL	P23	DP
DGND	P24	DG
DVDDL	P25	DP
DVDDL_CPU	P26	DP
DVDDL_CPU	P27	DP
DVDDL_CPU	P28	DP
DGND	P29	DG
MVDD	P30	MP
MVDDH	P31	MP
DGND	P32	DG
TP_P33	P33	
DGND	P34	DG

Pin Name	Pin No.	Type
DGND	P35	DG
DGND	P36	DG
ECC0	P37	I/O
ECC2	P38	I/O
SGND	R1	SG
SGND	R2	SG
HSO_N_S14	R3	AO
HSO_P_S14	R4	AO
SGND	R5	SG
SGND	R6	SG
SGND	R7	SG
HSIN_S14	R8	AI
HSIP_S14	R9	AI
SGND	R10	SG
SVDD_CK14	R11	SP
DGND	R12	DG
DVDDL	R13	DP
DGND	R14	DG
DVDDL	R15	DP
DGND	R16	DG
DVDDL	R17	DP
DGND	R18	DG
DVDDL	R19	DP
DGND	R20	DG
DVDDL	R21	DP
DGND	R22	DG
DVDDL	R23	DP
DGND	R24	DG
DVDDL	R25	DP
DGND	R26	DG
DVDDL_CPU	R27	DP
DGND	R28	DG
DGND	R29	DG
MVDD	R30	MP
DGND	R31	DG
DGND	R32	DG
ALERT1	R33	I/O
RSTN	R34	O
CSN	R35	O
TP_R36	R36	
TP_R37	R37	
	R38	
HSO_N_S15	T1	AO
HSO_P_S15	T2	AO



Pin Name	Pin No.	Type
SGND	T3	SG
SGND	T4	SG
SVDD_TRX	T5	SP
HSIN_S15	T6	AI
HSIP_S15	T7	AI
SGND	T8	SG
SGND	T9	SG
SVDD_CK15	T10	SP
SGND	T11	SG
DGND	T12	DG
DVDDL	T13	DP
DGND	T14	DG
DVDDL	T15	DP
DGND	T16	DG
DVDDL	T17	DP
DGND	T18	DG
DVDDL	T19	DP
DGND	T20	DG
DVDDL	T21	DP
DGND	T22	DG
DVDDL	T23	DP
DGND	T24	DG
DVDDL	T25	DP
DGND	T26	DG
DVDDL	T27	DP
DGND	T28	DG
MVDD	T29	MP
DGND	T30	DG
MVDD	T31	MP
DGND	T32	DG
DGND	T33	DG
RAS#/A16	T34	O
BG1	T35	O
DGND	T36	DG
CLK1B	T37	O
CLK1	T38	O
SGND	U1	SG
SGND	U2	SG
SGND	U3	SG
SGND	U4	SG
SGND	U5	SG
SGND	U6	SG
SGND	U7	SG
SGND	U8	SG

Pin Name	Pin No.	Type
SGND	U9	SG
SGND	U10	SG
SGND	U11	SG
DGND	U12	DG
DVDDL	U13	DP
DGND	U14	DG
DVDDL	U15	DP
DGND	U16	DG
DVDDL	U17	DP
DGND	U18	DG
DVDDL	U19	DP
DGND	U20	DG
DVDDL	U21	DP
DGND	U22	DG
DVDDL	U23	DP
DGND	U24	DG
DVDDL	U25	DP
DGND	U26	DG
DVDDL	U27	DP
DGND	U28	DG
DGND	U29	DG
MVDD	U30	MP
DGND	U31	DG
DGND	U32	DG
DGND	U33	DG
CKE1	U34	O
CAS#/A15	U35	O
BA1	U36	O
CLKB	U37	O
CLK	U38	O
HSOP_S16	V1	AO
HSOP_S16	V2	AO
SGND	V3	SG
HSIP_S16	V4	AI
HSIN_S16	V5	AI
SGND	V6	SG
SGND	V7	SG
SGND	V8	SG
SGND	V9	SG
DGND	V10	DG
DVDDL	V11	DP
DGND	V12	DG
DVDDL	V13	DP
DGND	V14	DG

Pin Name	Pin No.	Type
DVDDL	V15	DP
DGND	V16	DG
DVDDL	V17	DP
DGND	V18	DG
DVDDL	V19	DP
DGND	V20	DG
DVDDL	V21	DP
DGND	V22	DG
DVDDL	V23	DP
DGND	V24	DG
DVDDL	V25	DP
DGND	V26	DG
DVDDL	V27	DP
DGND	V28	DG
MVDD CK	V29	MP
DGND	V30	DG
MVDD	V31	MP
DGND	V32	DG
CS1N	V33	O
TEN	V34	O
DGND	V35	DG
A9	V36	O
CKE	V37	O
	V38	
SGND	W1	SG
SGND	W2	SG
SGND	W3	SG
SGND	W4	SG
SGND	W5	SG
SGND	W6	SG
SGND	W7	SG
SGND	W8	SG
SGND	W9	SG
DGND	W10	DG
DVDDL	W11	DP
DGND	W12	DG
DVDDL	W13	DP
DGND	W14	DG
DVDDL	W15	DP
DGND	W16	DG
DVDDL	W17	DP
DGND	W18	DG
DVDDL	W19	DP
DGND	W20	DG

Pin Name	Pin No.	Type
DVDDL	W21	DP
DGND	W22	DG
DVDDL	W23	DP
DGND	W24	DG
DVDDL	W25	DP
DGND	W26	DG
DVDDL	W27	DP
DGND	W28	DG
DGND	W29	DG
MVDD	W30	MP
DGND	W31	DG
DGND	W32	DG
DGND	W33	DG
DGND	W34	DG
A5	W35	O
ODT0	W36	O
A1	W37	O
A7	W38	O
HSOP_S17	Y1	AO
HSOP_S17	Y2	AO
SGND	Y3	SG
HSIP_S17	Y4	AI
HSIN_S17	Y5	AI
SGND	Y6	SG
SGND	Y7	SG
SGND	Y8	SG
SGND	Y9	SG
DGND	Y10	DG
DVDDL	Y11	DP
DGND	Y12	DG
DVDDL	Y13	DP
DGND	Y14	DG
DVDDL	Y15	DP
DGND	Y16	DG
DVDDL	Y17	DP
DGND	Y18	DG
DVDDL	Y19	DP
DGND	Y20	DG
DVDDL	Y21	DP
DGND	Y22	DG
DVDDL	Y23	DP
DGND	Y24	DG
DVDDL	Y25	DP
DGND	Y26	DG

Pin Name	Pin No.	Type
DVDDL	Y27	DP
DGND	Y28	DG
MVDD	Y29	MP
DGND	Y30	DG
MVDD	Y31	MP
DGND	Y32	DG
WE#/A14	Y33	O
ACTN	Y34	O
DGND	Y35	DG
A0	Y36	O
A12/BC#	Y37	O
	Y38	
SGND	AA1	SG
SGND	AA2	SG
SGND	AA3	SG
SGND	AA4	SG
SGND	AA5	SG
SGND	AA6	SG
SGND	AA7	SG
SGND	AA8	SG
SGND	AA9	SG
DGND	AA10	DG
DVDDL	AA11	DP
DGND	AA12	DG
DVDDL	AA13	DP
DGND	AA14	DG
DVDDL	AA15	DP
DGND	AA16	DG
DVDDL	AA17	DP
DGND	AA18	DG
DVDDL	AA19	DP
DGND	AA20	DG
DVDDL	AA21	DP
DGND	AA22	DG
DVDDL	AA23	DP
DGND	AA24	DG
DVDDL	AA25	DP
DGND	AA26	DG
DVDDL	AA27	DP
DGND	AA28	DG
DGND	AA29	DG
MVDD	AA30	MP
DGND	AA31	DG
DGND	AA32	DG

Pin Name	Pin No.	Type
DGND	AA33	DG
DGND	AA34	DG
A13	AA35	O
A11	AA36	O
A2	AA37	O
A10/AP	AA38	O
HSOP_S18	AB1	AO
HSOP_S18	AB2	AO
SGND	AB3	SG
HSIP_S18	AB4	AI
HSIN_S18	AB5	AI
SGND	AB6	SG
SGND	AB7	SG
SGND	AB8	SG
SGND	AB9	SG
DGND	AB10	DG
DVDDL	AB11	DP
DGND	AB12	DG
DVDDL	AB13	DP
DGND	AB14	DG
DVDDL	AB15	DP
DGND	AB16	DG
DVDDL	AB17	DP
DGND	AB18	DG
DVDDL	AB19	DP
DGND	AB20	DG
DVDDL	AB21	DP
DGND	AB22	DG
DVDDL	AB23	DP
DGND	AB24	DG
DVDDL	AB25	DP
DGND	AB26	DG
DVDDL	AB27	DP
DGND	AB28	DG
MVDD	AB29	MP
DGND	AB30	DG
MVDD	AB31	MP
DGND	AB32	DG
ALERT0	AB33	I/O
A4	AB34	O
DGND	AB35	DG
BG0	AB36	O
A3	AB37	O
	AB38	

Pin Name	Pin No.	Type
SGND	AC1	SG
SGND	AC2	SG
SGND	AC3	SG
SGND	AC4	SG
SGND	AC5	SG
SGND	AC6	SG
SGND	AC7	SG
SGND	AC8	SG
SGND	AC9	SG
DGND	AC10	DG
DVDDL	AC11	DP
DGND	AC12	DG
DVDDL	AC13	DP
DGND	AC14	DG
DVDDL	AC15	DP
DGND	AC16	DG
DVDDL	AC17	DP
DGND	AC18	DG
DVDDL	AC19	DP
DGND	AC20	DG
DVDDL	AC21	DP
DGND	AC22	DG
DVDDL	AC23	DP
DGND	AC24	DG
DVDDL	AC25	DP
DGND	AC26	DG
DVDDL	AC27	DP
DGND	AC28	DG
DGND	AC29	DG
DGND	AC30	DG
DGND	AC31	DG
DGND	AC32	DG
ODT1	AC33	O
ALERT2	AC34	I/O
A8	AC35	O
BA0	AC36	O
PAR	AC37	O
A6	AC38	O
HSOP_S19	AD1	AO
HSOP_S19	AD2	AO
SGND	AD3	SG
HSIP_S19	AD4	AI
HSIN_S19	AD5	AI
SGND	AD6	SG

Pin Name	Pin No.	Type
SGND	AD7	SG
SGND	AD8	SG
SGND	AD9	SG
DGND	AD10	DG
DVDDL	AD11	DP
DGND	AD12	DG
DVDDH	AD13	DP
DGND	AD14	DG
DVDDL	AD15	DP
DGND	AD16	DG
DVDDL	AD17	DP
DGND	AD18	DG
DVDDL	AD19	DP
DGND	AD20	DG
DVDDL	AD21	DP
DGND	AD22	DG
DVDDL	AD23	DP
DGND	AD24	DG
DVDDL	AD25	DP
DGND	AD26	DG
DVDDL	AD27	DP
DVDDH	AD28	DP
DGND	AD29	DG
DGND	AD30	DG
DGND	AD31	DG
TP_AD32	AD32	
TP_AD33	AD33	
GPIO [31]	AD34	I/O
DRAM_SEL	AD35	I
DGND	AD36	DG
DGND	AD37	DG
DGND	AD38	DG
SGND	AE1	SG
SGND	AE2	SG
SGND	AE3	SG
SGND	AE4	SG
SGND	AE5	SG
SGND	AE6	SG
SGND	AE7	SG
SGND	AE8	SG
SGND	AE9	SG
DGND	AE10	DG
DVDDL	AE11	DP
DGND	AE12	DG

Pin Name	Pin No.	Type
DVDDL	AE13	DP
DGND	AE14	DG
DVDDL	AE15	DP
DGND	AE16	DG
DVDDL	AE17	DP
DGND	AE18	DG
DVDDL	AE19	DP
DGND	AE20	DG
DVDDL	AE21	DP
DGND	AE22	DG
DVDDL	AE23	DP
DGND	AE24	DG
DVDDL	AE25	DP
DGND	AE26	DG
DVDDL	AE27	DP
DVDDH	AE28	DP
DGND	AE29	DG
DVDD_GPIO	AE30	DP
DGND	AE31	DG
DGND	AE32	DG
GPIO [26]	AE33	I/O
GPIO [27]	AE34	I/O
GPIO [28]	AE35	I/O
GPIO [29]	AE36	I/O
GPIO [30]	AE37	I/O
RESERVED_AE38	AE38	
HSOP_S20	AF1	AO
HSOP_S20	AF2	AO
SGND	AF3	SG
HSIP_S20	AF4	AI
HSIN_S20	AF5	AI
SGND	AF6	SG
SGND	AF7	SG
SGND	AF8	SG
SGND	AF9	SG
DGND	AF10	DG
DVDDL	AF11	DP
DGND	AF12	DG
DVDDL	AF13	DP
DGND	AF14	DG
DVDDL	AF15	DP
DGND	AF16	DG
DVDDL	AF17	DP
DGND	AF18	DG

Pin Name	Pin No.	Type
DVDDL	AF19	DP
DGND	AF20	DG
DVDDL	AF21	DP
DGND	AF22	DG
DVDDL	AF23	DP
DGND	AF24	DG
DVDDL	AF25	DP
DGND	AF26	DG
DVDDL	AF27	DP
DGND	AF28	DG
DGND	AF29	DG
DGND	AF30	DG
DVDD_JTAG	AF31	DP
DGND	AF32	DG
GPIO [23]	AF33	I/O
GPIO [24]	AF34	I/O
GPIO [25]	AF35	I/O
DDR_DATA_BUS_SEL	AF36	I
RESERVED_AF37	AF37	
RESERVED_AF38	AF38	
SGND	AG1	SG
SGND	AG2	SG
SGND	AG3	SG
SGND	AG4	SG
SGND	AG5	SG
SGND	AG6	SG
SGND	AG7	SG
SGND	AG8	SG
SGND	AG9	SG
SGND	AG10	SG
SGND	AG11	SG
SGND	AG12	SG
SGND	AG13	SG
SGND	AG14	SG
SGND	AG15	SG
SGND	AG16	SG
SGND	AG17	SG
SGND	AG18	SG
SGND	AG19	SG
SGND	AG20	SG
SGND	AG21	SG
SGND	AG22	SG
SGND	AG23	SG
SGND	AG24	SG

Pin Name	Pin No.	Type
SGND	AG25	SG
SGND	AG26	SG
SGND	AG27	SG
DGND	AG28	DG
DGND	AG29	DG
DVDD_SPI	AG30	DP
DGND	AG31	DG
DGND	AG32	DG
GPIO [21]	AG33	I/O
GPIO [22]	AG34	I/O
DGND	AG35	DG
GPIO [6] / JTAG_TDO	AG36	I/O
SYS_LED_EN	AG37	I
DDR_ECC_BUS_EN	AG38	I
HSOP_S21	AH1	AO
HSOP_S21	AH2	AO
SGND	AH3	SG
HSIP_S21	AH4	AI
HSIN_S21	AH5	AI
SGND	AH6	SG
SGND	AH7	SG
SGND	AH8	SG
SGND	AH9	SG
SGND	AH10	SG
SGND	AH11	SG
SGND	AH12	SG
SGND	AH13	SG
SGND	AH14	SG
SGND	AH15	SG
SGND	AH16	SG
SGND	AH17	SG
SGND	AH18	SG
SGND	AH19	SG
SGND	AH20	SG
SGND	AH21	SG
SGND	AH22	SG
SGND	AH23	SG
SGND	AH24	SG
SGND	AH25	SG
SGND	AH26	SG
SGND	AH27	SG
SGND	AH28	SG
DGND	AH29	DG
DGND	AH30	DG

Pin Name	Pin No.	Type
DVDD_MDX_3	AH31	DP
DGND	AH32	DG
GPIO [11] / MSPI_CS#1	AH33	I/O
GPIO [20]	AH34	I/O
GPIO [2] / JTAG_TMS	AH35	I/O
GPIO [3] / JTAG_TCK	AH36	I/O
GPIO [4] / JTAG_TRST#	AH37	I/O
GPIO [5] / JTAG_TDI	AH38	I/O
SGND	AJ1	SG
SGND	AJ2	SG
SGND	AJ3	SG
SGND	AJ4	SG
SGND	AJ5	SG
SGND	AJ6	SG
SGND	AJ7	SG
SGND	AJ8	SG
SGND	AJ9	SG
SGND	AJ10	SG
SGND	AJ11	SG
SGND	AJ12	SG
SGND	AJ13	SG
SGND	AJ14	SG
SVDDH_RX_25G	AJ15	SP
SGND	AJ16	SG
SVDDH_RX_25G	AJ17	SP
SGND	AJ18	SG
SGND	AJ19	SG
SGND	AJ20	SG
SVDDH_TX_25G	AJ21	SP
SGND	AJ22	SG
SGND	AJ23	SG
SGND	AJ24	SG
SVDDH_25G	AJ25	SP
SGND	AJ26	SG
SGND	AJ27	SG
SGND	AJ28	SG
DGND	AJ29	DG
DVDD_MDX_2	AJ30	DP
AGND	AJ31	AG
XTAL_AV09	AJ32	AP
DGND	AJ33	DG
GPIO [8] / MSPI_MISO	AJ34	I/O
GPIO [9] / MSPI_MOSI	AJ35	I/O
GPIO [10] / MSPI_CS#0	AJ36	I/O

Pin Name	Pin No.	Type
GPIO_MDC_B	AJ37	O
GPIO_MDIO_B	AJ38	I/O
HSOP_S22	AK1	AO
HSOP_S22	AK2	AO
SGND	AK3	SG
HSIP_S22	AK4	AI
HSIN_S22	AK5	AI
SGND	AK6	SG
SGND	AK7	SG
SGND	AK8	SG
SGND	AK9	SG
SVDDCK_HQ_G4_25G	AK10	SP
SGND	AK11	SG
SVDDCK_LQ_G4_25G	AK12	SP
SGND	AK13	SG
SVDDTRX_25G	AK14	SP
SGND	AK15	SG
SVDDTRX_25G	AK16	SP
SGND	AK17	SG
SVDDTRX_25G	AK18	SP
SGND	AK19	SG
SVDDTRX_25G	AK20	SP
SGND	AK21	SG
SVDDTRX_25G	AK22	SP
SGND	AK23	SG
SVDDTRX_25G	AK24	SP
SGND	AK25	SG
SVDDTRX_25G	AK26	SP
SGND	AK27	SG
SVDDTRX_25G	AK28	SP
DGND	AK29	DG
DGND	AK30	DG
AGND	AK31	AG
XTAL_AV33	AK32	AP
DGND	AK33	DG
GPIO [7] / MSPI_SCK	AK34	I/O
UART0_TXD	AK35	O
TP_AK36	AK36	
DGND	AK37	DG
TP_AK38	AK38	
SGND	AL1	SG
SGND	AL2	SG
SGND	AL3	SG
SGND	AL4	SG

Pin Name	Pin No.	Type
SGND	AL5	SG
SGND	AL6	SG
SGND	AL7	SG
SGND	AL8	SG
SGND	AL9	SG
SGND	AL10	SG
SGND	AL11	SG
SGND	AL12	SG
SGND	AL13	SG
SGND	AL14	SG
SGND	AL15	SG
SGND	AL16	SG
SGND	AL17	SG
SGND	AL18	SG
SGND	AL19	SG
SGND	AL20	SG
SGND	AL21	SG
SGND	AL22	SG
SGND	AL23	SG
SGND	AL24	SG
SGND	AL25	SG
SGND	AL26	SG
SGND	AL27	SG
SGND	AL28	SG
DGND	AL29	DG
AVDD09_PLL	AL30	AP
AGND	AL31	AG
AGND	AL32	AG
USB_LED	AL33	O
UART0_RXD	AL34	I
DGND	AL35	DG
TP_AL36	AL36	
TP_AL37	AL37	
TP_AL38	AL38	
HSOP_S23	AM1	AO
HSOP_S23	AM2	AO
SGND	AM3	SG
HSIP_S23	AM4	AI
HSIN_S23	AM5	AI
SGND	AM6	SG
SGND	AM7	SG
SGND	AM8	SG
SGND	AM9	SG
SVDDCK_HQ_G5_25G	AM10	SP

Pin Name	Pin No.	Type
SGND	AM11	SG
SVDDCK_LQ_G5_25G	AM12	SP
SGND	AM13	SG
SVDDCK_HQ_G6_25G	AM14	SP
SGND	AM15	SG
SVDDCK_LQ_G6_25G	AM16	SP
SGND	AM17	SG
SVDDCK_HQ_G7_25G	AM18	SP
SGND	AM19	SG
SVDDCK_LQ_G7_25G	AM20	SP
SGND	AM21	SG
SVDDCK_HQ_G8_25G	AM22	SP
SGND	AM23	SG
SVDDCK_LQ_G8_25G	AM24	SP
SGND	AM25	SG
SVDDCK_HQ_G9_25G	AM26	SP
SGND	AM27	SG
SVDDCK_LQ_G9_25G	AM28	SP
DGND	AM29	DG
AVDD33_PLL	AM30	AP
AGND	AM31	AG
AVDD33 (25G_BG)	AM32	AP
RESERVED_AM33	AM33	
UART0_CTS#	AM34	I
UART0_RTS#	AM35	O
SCL1	AM36	O
SDA1	AM37	I/O
DGND	AM38	DG
SGND	AN1	SG
SGND	AN2	SG
SGND	AN3	SG
SGND	AN4	SG
SGND	AN5	SG
SGND	AN6	SG
SGND	AN7	SG
SGND	AN8	SG
SGND	AN9	SG
SGND	AN10	SG
SGND	AN11	SG
SGND	AN12	SG
SGND	AN13	SG
SGND	AN14	SG
SGND	AN15	SG
SGND	AN16	SG

Pin Name	Pin No.	Type
SGND	AN17	SG
SGND	AN18	SG
SGND	AN19	SG
SGND	AN20	SG
SGND	AN21	SG
SGND	AN22	SG
SGND	AN23	SG
SGND	AN24	SG
SGND	AN25	SG
SGND	AN26	SG
SGND	AN27	SG
SGND	AN28	SG
SGND	AN29	SG
SGND	AN30	SG
SGND	AN31	SG
SGND	AN32	SG
PTP_CLK_IN	AN33	AI
RESERVED_AN34	AN34	
UART1_TXD	AN35	O
DGND	AN36	DG
SDA0	AN37	I/O
SCL0	AN38	O
HSIN_S24	AP1	AI
SGND	AP2	SG
HSIN_S25	AP3	AI
SGND	AP4	SG
HSIN_S26	AP5	AI
SGND	AP6	SG
HSIN_S27	AP7	AI
SGND	AP8	SG
HSIN_S28	AP9	AI
SGND	AP10	SG
HSIN_S29	AP11	AI
SGND	AP12	SG
HSIN_S30	AP13	AI
SGND	AP14	SG
HSIN_S31	AP15	AI
SGND	AP16	SG
HSIN_S32	AP17	AI
SGND	AP18	SG
HSIN_S33	AP19	AI
SGND	AP20	SG
HSIN_S34	AP21	AI
SGND	AP22	SG

Pin Name	Pin No.	Type
HSIN_S35	AP23	AI
SGND	AP24	SG
HSIN_S36	AP25	AI
SGND	AP26	SG
HSIN_S37	AP27	AI
SGND	AP28	SG
HSIN_S38	AP29	AI
SGND	AP30	SG
HSIN_S39	AP31	AI
SGND	AP32	SG
PTP_CLKN	AP33	AI
UART1_CTS#	AP34	I
UART1_RXD	AP35	I
MDIO_3	AP36	I/O
MDC_2	AP37	O
MDIO_2	AP38	I/O
HSIP_S24	AR1	AI
SGND	AR2	SG
HSIP_S25	AR3	AI
SGND	AR4	SG
HSIP_S26	AR5	AI
SGND	AR6	SG
HSIP_S27	AR7	AI
SGND	AR8	SG
HSIP_S28	AR9	AI
SGND	AR10	SG
HSIP_S29	AR11	AI
SGND	AR12	SG
HSIP_S30	AR13	AI
SGND	AR14	SG
HSIP_S31	AR15	AI
SGND	AR16	SG
HSIP_S32	AR17	AI
SGND	AR18	SG
HSIP_S33	AR19	AI
SGND	AR20	SG
HSIP_S34	AR21	AI
SGND	AR22	SG
HSIP_S35	AR23	AI
SGND	AR24	SG
HSIP_S36	AR25	AI
SGND	AR26	SG
HSIP_S37	AR27	AI
SGND	AR28	SG

Pin Name	Pin No.	Type
HSIP_S38	AR29	AI
SGND	AR30	SG
HSIP_S39	AR31	AI
SGND	AR32	SG
PTP_CLKP	AR33	AI
DGND	AR34	DG
UART1_RTS#	AR35	O
MDC_3	AR36	O
DGND	AR37	DG
RESERVED_AR38	AR38	
SGND	AT1	SG
SGND	AT2	SG
SGND	AT3	SG
SGND	AT4	SG
SGND	AT5	SG
SGND	AT6	SG
SGND	AT7	SG
SGND	AT8	SG
SGND	AT9	SG
SGND	AT10	SG
SGND	AT11	SG
SGND	AT12	SG
SGND	AT13	SG
SGND	AT14	SG
SGND	AT15	SG
SGND	AT16	SG
SGND	AT17	SG
SGND	AT18	SG
SGND	AT19	SG
SGND	AT20	SG
SGND	AT21	SG
SGND	AT22	SG
SGND	AT23	SG
SGND	AT24	SG
SGND	AT25	SG
SGND	AT26	SG
SGND	AT27	SG
SGND	AT28	SG
SGND	AT29	SG
SGND	AT30	SG
SGND	AT31	SG
SGND	AT32	SG
DGND	AT33	DG
PTP_CLK_S	AT34	I

Pin Name	Pin No.	Type
25G_REXT (IBREF)	AT35	AI/O
RESERVED_AT36	AT36	
TP_AT37	AT37	
TP_AT38	AT38	
SGND	AU1	SG
HSOP_S24	AU2	AO
SGND	AU3	SG
HSOP_S25	AU4	AO
SGND	AU5	SG
HSOP_S26	AU6	AO
SGND	AU7	SG
HSOP_S27	AU8	AO
SGND	AU9	SG
HSOP_S28	AU10	AO
SGND	AU11	SG
HSOP_S29	AU12	AO
SGND	AU13	SG
HSOP_S30	AU14	AO
SGND	AU15	SG
HSOP_S31	AU16	AO
SGND	AU17	SG
HSOP_S32	AU18	AO
SGND	AU19	SG
HSOP_S33	AU20	AO
SGND	AU21	SG
HSOP_S34	AU22	AO
SGND	AU23	SG
HSOP_S35	AU24	AO
SGND	AU25	SG
HSOP_S36	AU26	AO
SGND	AU27	SG
HSOP_S37	AU28	AO
SGND	AU29	SG
HSOP_S38	AU30	AO
SGND	AU31	SG
HSOP_S39	AU32	AO
SGND	AU33	SG
REFCLKN_156P25	AU34	AI
DGND	AU35	DG
BOOT_SEL[1]	AU36	I

Pin Name	Pin No.	Type
BOOT_SEL[0]	AU37	I
RESERVED_AU38	AU38	
SGND	AV1	SG
HSOP_S24	AV2	AO
SGND	AV3	SG
HSOP_S25	AV4	AO
SGND	AV5	SG
HSOP_S26	AV6	AO
SGND	AV7	SG
HSOP_S27	AV8	AO
SGND	AV9	SG
HSOP_S28	AV10	AO
SGND	AV11	SG
HSOP_S29	AV12	AO
SGND	AV13	SG
HSOP_S30	AV14	AO
SGND	AV15	SG
HSOP_S31	AV16	AO
SGND	AV17	SG
HSOP_S32	AV18	AO
SGND	AV19	SG
HSOP_S33	AV20	AO
SGND	AV21	SG
HSOP_S34	AV22	AO
SGND	AV23	SG
HSOP_S35	AV24	AO
SGND	AV25	SG
HSOP_S36	AV26	AO
SGND	AV27	SG
HSOP_S37	AV28	AO
SGND	AV29	SG
HSOP_S38	AV30	AO
SGND	AV31	SG
HSOP_S39	AV32	AO
SGND	AV33	SG
REFCLKP_156P25	AV34	AI
DGND	AV35	DG
XO	AV36	AO
XI	AV37	AI
DGND	AV38	DG

5. Pin Descriptions

5.1. DDR4 SDRAM Interface

Table 2. DDR4 SDRAM Interface Pins

Pin Name	Pin No.	Type	Description
DQ[31:0]	C37, D38, C36, D37, B38, E37, B37, E36, B35, C35, B34, D35, A35, C33, A34, D34, J37, K38, J36, K37, H38, L37, H37, L36, H35, J35, H34, K35, G35, J33, G34, K34	I/O	Data Input/Output: Bi-directional data bus. These pins must be floating when not used.
ECC[3:0]	N35, P38, N34, P37	I/O	ECC Data Input/Output: Bi-directional data bus. These pins must be floating when not used.
A[16:0]	T34, U35, Y33, AA35, Y37, AA36, AA38, V36, AC35, W38, AC38, W35, AB34, AB37, AA37, W37, Y36	O	Address Outputs.
BA[1:0]	U36, AC36	O	Bank Address Outputs.
DM3#/DBIU# DM2#/DBIL# DM1#/DBIU# DM0#/DBIL#	D36, A33, K36, G33	I/O	Data Masks and Data Bus Inversion. These pins must be floating when not used.
RAS#/A16, CAS#/A15, WEN#/A14	T34, U35, Y33	O	Command Outputs: RASN, CASN, WEN (along with CSN) define the command being entered.
CSN CS1N	R35 V33	O	Chip Select: CSN enables (registered LOW) and disables (registered HIGH) the command decoder. All commands are masked when CSN is registered HIGH.
CLK, CLKB	U38, U37	O	Clock: CLK and CLKB are differential clock outputs.
CLK1, CLK1B	T38, T37	O	Clock: CLK1 and CLK1B are differential clock outputs.
CKE	V37	O	Clock Enable: CKE HIGH activates.
CKE1	U34	O	Clock Enable: CKE1 HIGH activates.
DQS0, DQS0B, DQS1, DQS1B, DQS2, DQS2B, DQS3, DQS3B ECC_DQS0, ECC_DQS0B,	L35, L34, G37, G36, E35, E34, A37, A36 N37, N36	I/O	Data strobe. These pins must be floating when not used.
RSTN	R34	O	Active Low Reset.
ODT[1:0]	AC33, W36	O	On Die Termination: ODT (registered HIGH) enables termination resistance internal to the DDR4 SDRAM.

Pin Name	Pin No.	Type	Description
ACTN	Y34	O	Activation command output
BG[1:0]	T35, AB36	O	Bank group output
A10/AP	AA38	O	Auto-precharge
A12/BC#	Y37	O	Burst chop
PAR	AC37	O	Command and address parity output
ALERT0, ALERT1, ALERT2	AB33, R33, AC34	I/O	Alert: It has multi functions such as CRC error flag, command and address parity error flag. These pins must be connected to MVDD with 10K ohm resistor. These pins must be floating when not used.
TEN	V34	O	Connectivity test mode enable

5.2. SPI Flash Controller Interface

Table 3. SPI Flash Controller Interface Pins

Pin Name	Pin No.	Type	Description
SPI_CS#[1:0]	B22, D22	O	Chip select 1-0.
SPI_SCLK	D21	O	Clock output.
SPI_SDI/SIO0	E22	I/O	Serial data output (for 1Xi/O)/ Serial data input & output (for 2Xi/O or 4Xi/O). This pin must be floating when not used.
SPI_SDO/SIO1	A21	I/O	Serial data input (for 1Xi/O)/ Serial data input & output (for 2Xi/O or 4Xi/O). This pin must be floating when not used.
SPI_SIO2	B21	I/O	Serial data input & output (for 4Xi/O). This pin must be floating when not used.
SPI_SIO3	C21	I/O	Serial data input & output (for 4Xi/O). This pin must be floating when not used.
SPI_RSTN	E21	O	Hardware Reset output (active low)

5.3. Parallel NAND Flash Controller Interface

Table 4. Parallel NAND Flash Controller Interface Pins

Pin Name	Pin No.	Type	Description
NAND_CE#[1:0]	A25, B25	O	Chip select 1-0.
NAND_RB_N	C26	I	Ready/Busy This pin must be floating when not used.
NAND_RE_N	C25	O	Read enable
NAND_WE_N	A24	O	Write enable
NAND_CLE	D25	O	Command latch enable
NAND_ALE	E25	O	Address latch enable
NAND_IO[7:0]	D23, A22, D24, C24, E24, C23, B23, A23	I/O	Data Input/Output These pins must be floating when not used.
NAND_WP_N	F24	O	Write protect

5.4. eMMC Flash Controller Interface

Table 5. eMMC Flash Controller Interface Pins

Pin Name	Pin No.	Type	Description
SD_CLK	B25	O	Clock output
SD_CMD	A25	I/O	Command This pin must be floating when not used.
SD_DS	C26	I	Data Strobe This pin must be floating when not used.
SD_DAT[7:0]	D23, A22, D24, C24, E24, C23, B23, A23	I/O	Data Input/Output These pins must be floating when not used.
SD_RST_N	E25	O	Reset

5.5. I2C Interface

Table 6. I2C Interface Pins

Pin Name	Pin No.	Type	Description
SCL0	AN38	O	Master I2C 0 Serial Clock
SDA0	AN37	I/O	Master I2C 0 Serial Data/Address This pin must be floating when not used.
SCL1	AM36	O	Master I2C 1 Serial Clock
SDA1	AM37	I/O	Master I2C 1 Serial Data/Address This pin must be floating when not used.
SCL2	C17	O	Master I2C 2 Serial Clock
SDA2	D16	I/O	Master I2C 2 Serial Data/Address This pin must be floating when not used.

Pin Name	Pin No.	Type	Description
SCL3	B15	O	Master I2C 3 Serial Clock
SDA3	C15	I/O	Master I2C 3 Serial Data/Address This pin must be floating when not used.

5.6. UART Interface

Table 7. UART Interface Pins

Pin Name	Pin No.	Type	Description
UART0_TXD	AK35	O	UART0 Interface Transmit Data.
UART0_RXD	AL34	I	UART0 Interface Receive Data. This pin must be floating when not used.
UART0_RTS#	AM35	O	UART0 Interface Request to Send.
UART0_CTS#	AM34	I	UART0 Interface Clear to Send. This pin must be floating when not used.
UART1_TXD	AN35	O	UART1 Interface Transmit Data.
UART1_RXD	AP35	I	UART1 Interface Receive Data. This pin must be floating when not used.
UART1_RTS#	AR35	O	UART1 Interface Request to Send.
UART1_CTS#	AP34	I	UART1 Interface Clear to Send. This pin must be floating when not used.
UART2_TXD	E20	O	UART2 Interface Transmit Data.
UART2_RXD	F21	I	UART2 Interface Receive Data. This pin must be floating when not used.

5.7. EJTAG Interface

Table 8. EJTAG Interface Pins

Pin Name	Pin No.	Type	Description
JTAG_TMS	AH35	I	JTAG Test Mode Select. This pin must be floating when not used.
JTAG_TCK	AH36	I	JTAG Test Clock Input. This pin must be floating when not used.
JTAG_TRST#	AH37	I	JTAG Test Reset. <i>Note: The pin must be external pulled down via a 1Kohm resistor.</i> This pin must be floating when not used.
JTAG_TDI	AH38	I	JTAG Test Data Input. This pin must be floating when not used.
JTAG_TDO	AG36	O	JTAG Test Data Output.

5.8. GPIO Interface

Table 9. GPIO Interface Pins

Pin Name	Pin No.	Type	Description
GPIO[31:0]	AD34, AE37, AE36, AE35, AE34, AE33, AF35, AF34, AF33, AG34, AG33, AH34, F18, E20, F21, G22, F22, G23, F23, G25, AH33, AJ36, AJ35, AJ34, AK34, AG36, AH38, AH37, AH36, AH35, E18, G19	I/O	General Purpose Input/Output. Provides configurable I/O ports that can be configured for either input or output. These pins must be floating when not used. GPIO[4] cannot connect with internal pull high device.

5.9. 10G Serdes Interface

Table 10. 10G Serdes Interface Pins

Pin Name	Pin No.	Type	Description
HSOP_S[15:0]/ HSOP_N[15:0]	T2, R4, P2, N4, M2, L4, K2, J4, H2, G4, F2, E4, D2, C4, B2, A4 T1, R3, P1, N3, M1, L3, K1, J3, H1, G3, F1, E3, D1, C3, B1, A3	AO	10G Serdes (up to 10.3125G) Transmit Data Differential Output Pair.
HSIP_S[15:0]/ HSIN_S[15:0]	T7, R9, P7, N9, M7, L9, K7, J9, H7, G9, F7, E9, D7, C9, B7, A9 T6, R8, P6, N8, M6, L8, K6, J8, H6, G8, F6, E8, D6, C8, B6, A8	AI	10G Serdes (up to 10.3125G) Receive Data Differential Input Pair. These pins must be connected with a series 1K ohm resistor to ground when not used.

5.10. 25G Serdes Interface

Table 11. 25G Serdes Interface Pins

Pin Name	Pin No.	Type	Description
HSOP_S[39:16]/ HSOP_N[39:16]	AV32, AV30, AV28, AV26, AV24, AV22, AV20, AV18, AV16, AV14, AV12, AV10, AV8, AV6, AV4, AV2, AM1, AK1, AH1, AF1, AD1, AB1, Y1, V1 AU32, AU30, AU28, AU26, AU24, AU22, AU20, AU18, AU16, AU14, AU12, AU10, AU8, AU6, AU4, AU2, AM2, AK2, AH2, AF2, AD2, AB2, Y2, V2	AO	25G Serdes (up to 25.78125G) Transmit Data Differential Output Pair.

Pin Name	Pin No.	Type	Description
HSIP_S[39:16]/ HSIN_S[39:16]	AR31, AR29, AR27, AR25, AR23, AR21, AR19, AR17, AR15, AR13, AR11, AR9, AR7, AR5, AR3, AR1, AM4, AK4, AH4, AF4, AD4, AB4, Y4, V4 AP31, AP29, AP27, AP25, AP23, AP21, AP19, AP17, AP15, AP13, AP11, AP9, AP7, AP5, AP3, AP1, AM5, AK5, AH5, AF5, AD5, AB5, Y5, V5	AI	25G Serdes (up to 25.78125G) Receive Data Differential Input Pair. These pins must be connected with a series 1K ohm resistor to ground when not used.

5.11. SGMII Interface

Table 12. SGMII Interface Pins

Pin Name	Pin No.	Type	Description
SGMII_HSOP SGMII_HSON	A31 B31	AO	SGMII (1.25G) Transmit Data Differential Output Pair.
SGMII_HSIP SGMII_HSIN	D31 E31	AI	SGMII (1.25G) Receive Data Differential Input Pair. These pins must be connected with a series 1K ohm resistor to ground when not used.

5.12. USB3.0 Interface

Table 13. USB3.0 Interface Pins

Pin Name	Pin No.	Type	Description
USB_HSOP USB_HSON	A27 B27	AO	USB3.0 Transmit Data Differential Output Pair.
USB_HSIP USB_HSIN	A29 B29	AI	USB3.0 Receive Data Differential Input Pair. These pins must be connected with a series 1K ohm resistor to ground when not used.

5.13. USB2.0 Interface

Table 14. USB2.0 Interface Pins

Pin Name	Pin No.	Type	Description
USB_HSDM USB_HSDP	D29 E29	AI/O	USB Data Differential input/output Pair. These pins must be floating when not used.

5.14. PCIE Interface

Table 15. PCIE Interface Pins

Pin Name	Pin No.	Type	Description
PCIE_HSOP[1:0] PCIE_HSON[1:0]	G31, G27 H31, H27	AO	PCIE Transmit Data Differential Output Pair.
PCIE_HSIP[1:0] PCIE_HSIN[1:0]	G29, J28 H29, K28	AI	PCIE Receiver Data Differential Input Pair. These pins must be pulled down with a 1K ohm resistor when not used.
PCIE_CLK_P PCIE_CLK_N	D27 E27	AI/O	PCIE Differential Clock. (100MHz) These pins must be pulled down with a 1K ohm resistor when not used.
PCIE_RSTB	F19	I/O	PCIE Reset. This pin must be floating when not used.

5.15. SyncE and PTP Interface

Table 16. SyncE and PTP Interface Pins

Pin Name	Pin No.	Type	Description
SYNC_OUT_CLK	A16	O	Primary recovered clock
SYNC_OUT_LOCK	B16	O	Indication the SYNC_OUT_CLK is valid.
SYNC_OUT_CLK_BKUP	A14	O	Secondary recovered clock
SYNC_OUT_LOCK_BKUP	B14	O	Indication the SYNC_OUT_CLK_BKUP is valid.
PTP_CLKP	AR33	AI	PTP Differential clock input This pin must be pulled down with a 1K ohm resistor when not used.
PTP_CLKN	AP33	AI	PTP Differential clock input This must be pulled up with a 10K ohm resistor when not used.
PTP_CLK_IN	AN33	AI	PTP Single-end clock input This pin must be pulled down with a 1K ohm resistor when not used.

5.16. LED Interface

Table 17. LED Interface Pins

Pin Name	Pin No.	Type	Description
LED_CLK	D18	O	Shift register clock output.
LED_DATA	B17	O	Serial data output.
LED_SYNC	E18	O	Storage register clock output.
LED_MDC	D18	O	Controller Clock for scan LED Mode.

Pin Name	Pin No.	Type	Description
LED_MDIO	B17	I/O	Controller Data for scan LED Mode. This pin must be floating when not used.
LED_STACK	F18	I/O	For Master and Slave LED sync This pin must be floating when not used.
SYS_LED	G19	O	System LED.
USB_LED	AL33	O	USB LED.

5.17. SPI Master Interface

Table 18. SPI Master Interface Pins

Pin Name	Pin No.	Type	Description
MSPI_SCK	AK34	O	SPI master, clock output.
MSPI_CS#0	AJ36	O	SPI master, chip select 0.
MSPI_CS#1	AH33	O	SPI master, chip select 1.
MSPI_MOSI	AJ35	O	SPI master, data output.
MSPI_MISO	AJ34	I	SPI master, data input. This pin must be floating when not used.

5.18. Miscellaneous Interface

Table 19. Miscellaneous Interface Pins

Pin Name	Pin No.	Type	Description
MDC_0	A18	O	MII 0 Management Interface Clock Pin.
MDIO_0	B18	I/O	MII 0 Management Interface Data Pin. This pin must be floating when not used.
MDC_1	A19	O	MII 1 Management Interface Clock Pin.
MDIO_1	B19	I/O	MII 1 Management Interface Data Pin. This pin must be floating when not used.
MDC_2	AP37	O	MII 2 Management Interface Clock Pin.
MDIO_2	AP38	I/O	MII 2 Management Interface Data Pin. This pin must be floating when not used.
MDC_3	AR36	O	MII 3 Management Interface Clock Pin.
MDIO_3	AP36	I/O	MII 3 Management Interface Data Pin. This pin must be floating when not used.
MDC_OOB	B20	O	MII for OOB Management Interface Clock Pin.
MDIO_OOB	C20	I/O	MII for OOB Management Interface Data Pin. This pin must be floating when not used.
GPIO_MDC_A	C19	O	GPIO A Management Interface Clock for RTL8231.
GPIO_MDIO_A	D19	I/O	GPIO A Management Interface Data for RTL8231. This pin must be floating when not used.
GPIO_MDC_B	AJ37	O	GPIO B Management Interface Clock for RTL8231.
GPIO_MDIO_B	AJ38	I/O	GPIO B Management Interface Data for RTL8231. This pin must be floating when not used.
XI	AV37	AI	25MHz Crystal Clock Input Pin.
XO	AV36	AO	25MHz Crystal Clock Output Pin.
REFCLKP_156P25 REFCLKN_156P25	AV34 AU34	AI	156.25MHz Differential Clock Input Pair.
RESETB	C14	AI	System Pin Reset Input.
REXT (IBREF) 25G_REXT (IBREF)	B12 AT35	AI/O	Reference Resistor for BG. A 2.49K Ω (1%) resistor should be connected between IBREF and GND.
PWRMON	D14	AI	For power monitor use. This pin must be floating when not used.
RST_OUT	F15	O	For reset other device.

Pin Name	Pin No.	Type	Description
RESERVED_A13	A13		Reserved. Must be pulled down with a 1K ohm resistor in normal operation.
RESERVED_B13	B13		
RESERVED_C13	C13		
RESERVED_D17	D17		
RESERVED_E15	E15		
RESERVED_E17	E17		
RESERVED_F14	F14		
RESERVED_F16	F16		
RESERVED_F17	F17		
RESERVED_F20	F20		
RESERVED_G16	G16		
RESERVED_G18	G18		
RESERVED_AE38	AE38		
RESERVED_AF37	AF37		
RESERVED_AF38	AF38		
RESERVED_AM33	AM33		
RESERVED_D12	D12		Reserved. Must be pulled up with a 10K ohm resistor in normal operation.
RESERVED_D13	D13		
RESERVED_E13	E13		
RESERVED_F12	F12		
RESERVED_F13	F13		
RESERVED_G13	G13		
RESERVED_G14	G14		
RESERVED_G15	G15		
RESERVED_AN34	AN34		
RESERVED_AT36	AT36		
TP_M30	M30		Reserved for test point in normal operation.
TP_M31	M31		
TP_AD32	AD32		
TP_AD33	AD33		
TP_AL38	AL38		
TP_AT37	AT37		
TP_AT38	AT38		
TP_R37	R37		
TP_R36	R36		
TP_L33	L33		
TP_E33	E33		
TP_P33	P33		
TP_AK36	AK36		
TP_AK38	AK38		
TP_AL36	AL36		
TP_AL37	AL37		
RESERVED_AU38	AU38		Reserved. Must be connected to DVDDL.
RESERVED_AR38	AR38		Reserved. Must be connected to DGND

5.19. Configuration (Pin Strapping Function)

Table 20. Configuration Strapping Pins

Pin Name	Pin No.	Default	Description
SYS_LED_EN	AG37	0b1	When enable System LED, the GPIO[0] change to system LED pin. 0b0: Disable system LED 0b1: Enable system LED
SOC_PCIE_MODE_SEL[1:0]	E16, G17	0b01	Enable CPU idle mode 0x0 : Internal CPU only; SoC on; PCIe clk and reset gating by reg 0x1 : EP mode; SoC power down; PCIe EP 0x2 : RC mode; SoC on; PCIe RC 0x3 : EP mode → Dual(External + Internal) CPU mode; SoC on; PCIe EP
PHY_SYS_STOP_PCIECK	G20	0b1	PCIe ref CLK input/output configuration 0x0: ref clk is output as RC mode 0x1: ref clk is input as EP mode
DDR_ECC_BUS_EN	AG38	0b0	Select DDR interface with ECC correction 0b0 : Without ECC 0b1 : With ECC
BOOT_SEL[1:0]	AU36, AU37	0b00	Boot selection 0x0: Serial NOR boot 0x1: Serial NAND boot 0x2: Parallel NAND boot 0x3: eMMC boot
DDR_DATA_BUS_SEL	AF36	0b0	Select DDR data bus width. 0b0: DQ16 0b1: DQ32
DDR_RANK_SEL	G21	0b1	Select DDR single rank/ dual rank 0b0 : Single rank 0b1 : Dual rank
HW_RESET_EN	D15	0b1	Triggering hardware reset signal or not before booting from SPI flash. 0b0: Don't trigger hardware reset signal 0b1: Trigger hardware reset signal
PTP_CLK_S	AT34	0b1	PTP clock source selection 0x0: Differential clk source 0x1: Single-end clk source
DRAM_SEL	AD35	0b0	Use 8-bit or 16-bit DRAM 0b0: 8-bit DRAM 0b1: 16-bit DRAM

5.20. Power and GND

Table 21. Power and GND Pins

Pin Name	Pin No.	Type	Description
SVDD_TRX	B5, D5, F5, H5, K5, M5, P5, T5	SP	10G SERDES TX/RX Low Voltage Power.
SVDD_CK0	A11	SP	10G SERDES CK/CMU Low Voltage Power.
SVDD_CK1	B10		
SVDD_CK2	C11		
SVDD_CK3	D10		
SVDD_CK4	E11		
SVDD_CK5	F10		
SVDD_CK6	G11		
SVDD_CK7	H10		
SVDD_CK8	J11		
SVDD_CK9	K10		
SVDD_CK10	L11		
SVDD_CK11	M10		
SVDD_CK12	N11		
SVDD_CK13	P10		
SVDD_CK14	R11		
SVDD_CK15	T10		
SVDDTRX_25G	AK14, AK16, AK18, AK20, AK22, AK24, AK26, AK28	SP	25G SERDES TX/RX Low Voltage Power.
SVDDCK_HQ_G4_25G	AK10	SP	25G SERDES CK/CMU HQ Voltage Power.
SVDDCK_HQ_G5_25G	AM10		
SVDDCK_HQ_G6_25G	AM14		
SVDDCK_HQ_G7_25G	AM18		
SVDDCK_HQ_G8_25G	AM22		
SVDDCK_HQ_G9_25G	AM26		
SVDDCK_LQ_G4_25G	AK12	SP	25G SERDES CK/CMU LQ Voltage Power.
SVDDCK_LQ_G5_25G	AM12		
SVDDCK_LQ_G6_25G	AM16		
SVDDCK_LQ_G7_25G	AM20		
SVDDCK_LQ_G8_25G	AM24		
SVDDCK_LQ_G9_25G	AM28		
SVDDH_25G	AJ25	SP	25G SERDES PLL/BG (General) High Voltage Power.
SVDDH_TX_25G	AJ21	SP	25G SERDES TX LDO using Voltage Power.
SVDDH_RX_25G	AJ15, AJ17	SP	25G SERDES RX LDO using Voltage Power.
PCIE_VDDL	F30, J30	AP	PCIE TX/RX Low Voltage Power.
PCIE_CK	F26	AP	PCIE CK/CMU Low Voltage Power.
USB_VDDL	C28	AP	USB Low Voltage Power.
USB_VDDH	F28	AP	USB High Voltage Power.
SVDDL_OOB_TRX	C30	AP	SGMII TX/RX Low Voltage Power.

Pin Name	Pin No.	Type	Description
MVDDH	P31	MP	DDR Controller High Voltage Power.
MVDD	T29, Y29, AB29, P30, R30, U30, W30, AA30, T31, V31, Y31, AB31	MP	DDR Controller Low Voltage Power.
MVDD_CK	V29	MP	DDR Controller CK Low Voltage Power.
AVDDV09 (BG)	H14	AP	BG for 0.9V Voltage Power.
AVDDV10 (BG)	H13	AP	BG for 1.0V Voltage Power.
AVDDH (BG)	H12	AP	BG High Voltage Power.
AVDD33 (25G_BG)	AM32	AP	25G BG High Voltage Power.
XTAL_AV09	AJ32	AP	XTAL Low Voltage Power.
XTAL_AV33	AK32	AP	XTAL High Voltage Power.
AVDD09_PLL	AL30	AP	PLL Low Voltage Power.
AVDD33_PLL	AM30	AP	PLL High Voltage Power.
AVDD09_PLL_CPU	L30	AP	PLL for CPU Low Voltage Power.
AVDD33_PLL_CPU	L31	AP	PLL for CPU High Voltage Power.
DVDD_MDX_0	J21	DP	Digital Power for MDX_0
DVDD_MDX_1	J22	DP	Digital Power for MDX_1
DVDD_MDX_2	AJ30	DP	Digital Power for MDX_2
DVDD_MDX_3	AH31	DP	Digital Power for MDX_3
DVDD_MDX_OOB	J23	DP	Digital Power for MDX_OOB
DVDD_NF	K22, K23	DP	Digital Power for SPI NOR & NAND Flash
DVDD_P	H25, J25	DP	Digital Power for Parallel NAND Flash & eMMC
DVDD_IO	J26, K26	DP	Digital Power for Parallel NAND Flash & eMMC data bus
DVDD_JTAG	AF31	DP	Digital Power for JTAG and Group_1 I/O Pins.
DVDD_SPI	AG30	DP	Digital Power for SPI_MST and Group_2 I/O Pins.
DVDD_GPIO	AE30	DP	Digital Power for GPIO[31:20] Pins.
DVDDH	K17, K18, K19, K20, AD13, AD28, AE28, M29	DP	Digital Power for I/O Pins.



Pin Name	Pin No.	Type	Description
DVDDL	V11, W11, Y11, AA11, AB11, AC11, AD11, AE11, AF11, M13, N13, P13, R13, T13, U13, V13, W13, Y13, AA13, AB13, AC13, AE13, AF13, M15, N15, P15, R15, T15, U15, V15, W15, Y15, AA15, AB15, AC15, AD15, AE15, AF15, M17, N17, P17, R17, T17, U17, V17, W17, Y17, AA17, AB17, AC17, AD17, AE17, AF17, M19, N19, P19, R19, T19, U19, V19, W19, Y19, AA19, AB19, AC19, AD19, AE19, AF19, M21, N21, P21, R21, T21, U21, V21, W21, Y21, AA21, AB21, AC21, AD21, AE21, AF21, M23, N23, P23, R23, T23, U23, V23, W23, Y23, AA23, AB23, AC23, AD23, AE23, AF23, M25, N25, P25, R25, T25, U25, V25, W25, Y25, AA25, AB25, AC25, AD25, AE25, AF25, M27, T27, U27, V27, W27, Y27, AA27, AB27, AC27, AD27, AE27, AF27	DP	Digital Low Power for Digital Core.
DVDDL CPU	P26, N27, P27, R27, N28, P28	DP	Digital Low Power for CPU.



DGND	V10, W10, Y10, AA10, AB10, AC10, AD10, AE10, AF10, A12, C12, E12, G12, K12, L12, M12, N12, P12, R12, T12, U12, V12, W12, Y12, AA12, AB12, AC12, AD12, AE12, AF12, K13, L13, E14, K14, L14, M14, N14, P14, R14, T14, U14, V14, W14, Y14, AA14, AB14, AC14, AD14, AE14, AF14, A15, H15, J15, K15, L15, C16, H16, J16, K16, L16, M16, N16, P16, R16, T16, U16, V16, W16, Y16, AA16, AB16, AC16, AD16, AE16, AF16, A17, H17, J17, L17, C18, H18, J18, L18, M18, N18, P18, R18, T18, U18, V18, W18, Y18, AA18, AB18, AC18, AD18, AE18, AF18, E19, H19, J19, L19, A20, D20, H20, J20, L20, M20, N20, P20, R20, T20, U20, V20, W20, Y20, AA20, AB20, AC20, AD20, AE20, AF20, H21, K21, L21, C22, H22, L22, M22, N22, P22, R22, T22, U22, V22, W22, Y22, AA22, AB22, AC22, AD22, AE22, AF22, E23, H23, L23, B24, G24, H24, J24, K24, L24, M24, N24, P24, R24, T24, U24, V24, W24, Y24, AA24, AB24, AC24, AD24, AE24, AF24, F25, K25, L25, L26, M26, N26, R26, T26, U26, V26, W26, Y26, AA26, AB26, AC26, AD26, AE26, AF26, L27, M28, R28, T28, U28, V28, W28, Y28, AA28, AB28, AC28, AF28, AG28, N29, P29, R29, U29, W29, AA29, AC29, AD29, AE29, AF29, AG29, AH29, AJ29, AK29, AL29, AM29, N30, T30, V30, Y30, AB30, AC30, AD30, AF30, AH30, AK30, N31, R31, U31, W31, AA31, AC31, AD31, AE31, AG31, A32, B32, C32, D32, E32, F32, G32, H32, J32, K32, L32, M32, N32, P32, R32, T32, U32, V32, W32, Y32, AA32, AB32, AC32, AE32, AF32, AG32, AH32, B33, D33, F33, H33, K33, M33, N33, T33, U33, W33, AA33, AJ33, AK33, AT33, C34, F34, J34, M34,	DG	Digital Ground.
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Pin Name	Pin No.	Type	Description
	P34, W34, AA34, AR34, F35, M35, P35, V35, Y35, AB35, AG35, AL35, AU35, AV35, B36, F36, H36, M36, P36, T36, AD36, AN36, F37, M37, AD37, AK37, AR37, F38, M38, AD38, AM38, AV38		



SGND	A1, C1, E1, G1, J1, L1, N1, R1, U1, W1, AA1, AC1, AE1, AG1, AJ1, AL1, AN1, AT1, AU1, AV1, A2, C2, E2, G2, J2, L2, N2, R2, U2, W2, AA2, AC2, AE2, AG2, AJ2, AL2, AN2, AP2, AR2, AT2, B3, D3, F3, H3, K3, M3, P3, T3, U3, V3, W3, Y3, AA3, AB3, AC3, AD3, AE3, AF3, AG3, AH3, AJ3, AK3, AL3, AM3, AN3, AT3, AU3, AV3, B4, D4, F4, H4, K4, M4, P4, T4, U4, W4, AA4, AC4, AE4, AG4, AJ4, AL4, AN4, AP4, AR4, AT4, A5, C5, E5, G5, J5, L5, N5, R5, U5, W5, AA5, AC5, AE5, AG5, AJ5, AL5, AN5, AT5, AU5, AV5, A6, C6, E6, G6, J6, L6, N6, R6, U6, V6, W6, Y6, AA6, AB6, AC6, AD6, AE6, AF6, AG6, AH6, AJ6, AK6, AL6, AM6, AN6, AP6, AR6, AT6, A7, C7, E7, G7, J7, L7, N7, R7, U7, V7, W7, Y7, AA7, AB7, AC7, AD7, AE7, AF7, AG7, AH7, AJ7, AK7, AL7, AM7, AN7, AT7, AU7, AV7, B8, D8, F8, H8, K8, M8, P8, T8, U8, V8, W8, Y8, AA8, AB8, AC8, AD8, AE8, AF8, AG8, AH8, AJ8, AK8, AL8, AM8, AN8, AP8, AR8, AT8, B9, D9, F9, H9, K9, M9, P9, T9, U9, V9, W9, Y9, AA9, AB9, AC9, AD9, AE9, AF9, AG9, AH9, AJ9, AK9, AL9, AM9, AN9, AT9, AU9, AV9, A10, C10, E10, G10, J10, L10, N10, R10, U10, AG10, AH10, AJ10, AL10, AN10, AP10, AR10, AT10, B11, D11, F11, H11, K11, M11, P11, T11, U11, AG11, AH11, AJ11, AK11, AL11, AM11, AN11, AT11, AU11, AV11, AG12, AH12, AJ12, AL12, AN12, AP12, AR12, AT12, AG13, AH13, AJ13, AK13, AL13, AM13, AN13, AT13, AU13, AV13, AG14, AH14, AJ14, AL14, AN14, AP14, AR14, AT14, AG15, AH15, AK15, AL15, AM15, AN15, AT15, AU15, AV15, AG16, AH16, AJ16, AL16, AN16,	SG	SERDES Ground.
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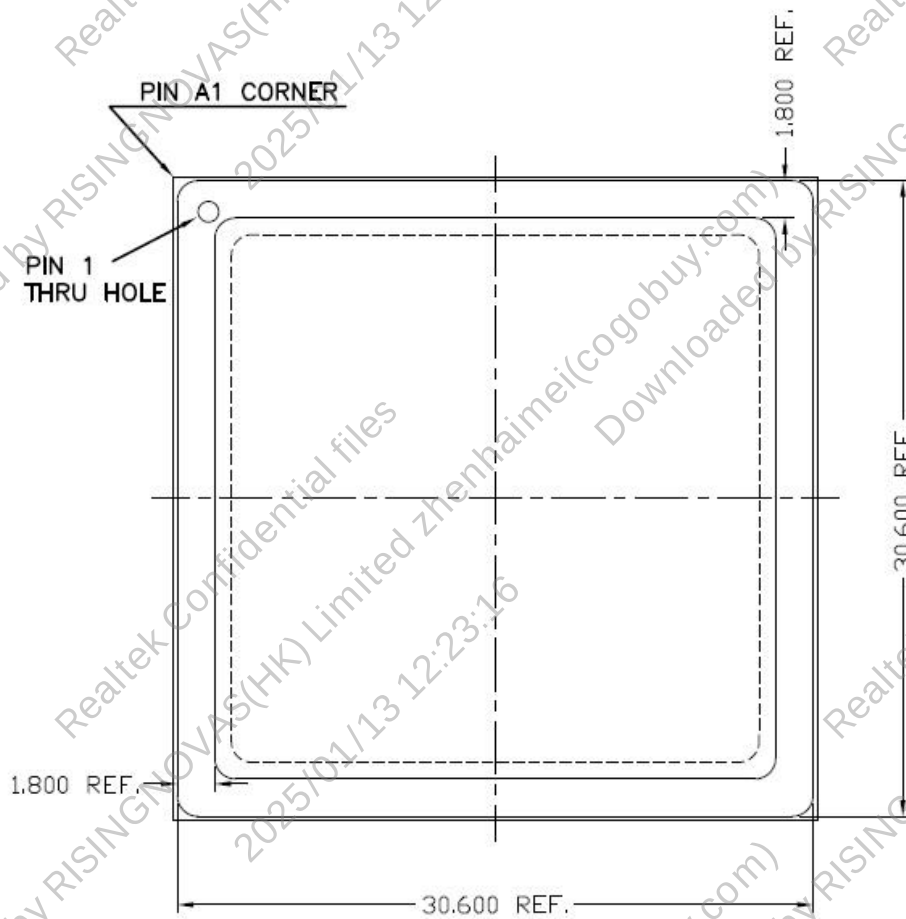


Pin Name	Pin No.	Type	Description
	AP16, AR16, AT16, AG17, AH17, AK17, AL17, AM17, AN17, AT17, AU17, AV17, AG18, AH18, AJ18, AL18, AN18, AP18, AR18, AT18, AG19, AH19, AJ19, AK19, AL19, AM19, AN19, AT19, AU19, AV19, AG20, AH20, AJ20, AL20, AN20, AP20, AR20, AT20, AG21, AH21, AK21, AL21, AM21, AN21, AT21, AU21, AV21, AG22, AH22, AJ22, AL22, AN22, AP22, AR22, AT22, AG23, AH23, AJ23, AK23, AL23, AM23, AN23, AT23, AU23, AV23, AG24, AH24, AJ24, AL24, AN24, AP24, AR24, AT24, AG25, AH25, AK25, AL25, AM25, AN25, AT25, AU25, AV25, AG26, AH26, AJ26, AL26, AN26, AP26, AR26, AT26, AG27, AH27, AJ27, AK27, AL27, AM27, AN27, AT27, AU27, AV27, AH28, AJ28, AL28, AN28, AP28, AR28, AT28, AN29, AT29, AU29, AV29, AN30, AP30, AR30, AT30, AN31, AT31, AU31, AV31, AN32, AP32, AR32, AT32, AU33, AV33		
AGND	J12, J13, J14, A26, B26, D26, E26, G26, H26, C27, F27, J27, K27, A28, B28, D28, E28, G28, H28, L28, C29, F29, J29, K29, L29, G30, H30, K30, J31, K31, AJ31, AK31, AL31, AM31, AL32, A30, B30, D30, E30, C31, F31	AG	Analog Ground

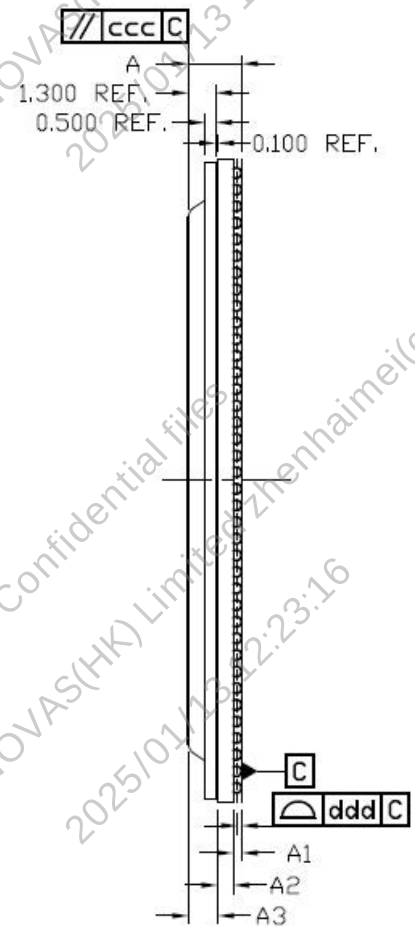
6. Package Information

6.1. HFC FBGA1433 (31*31)

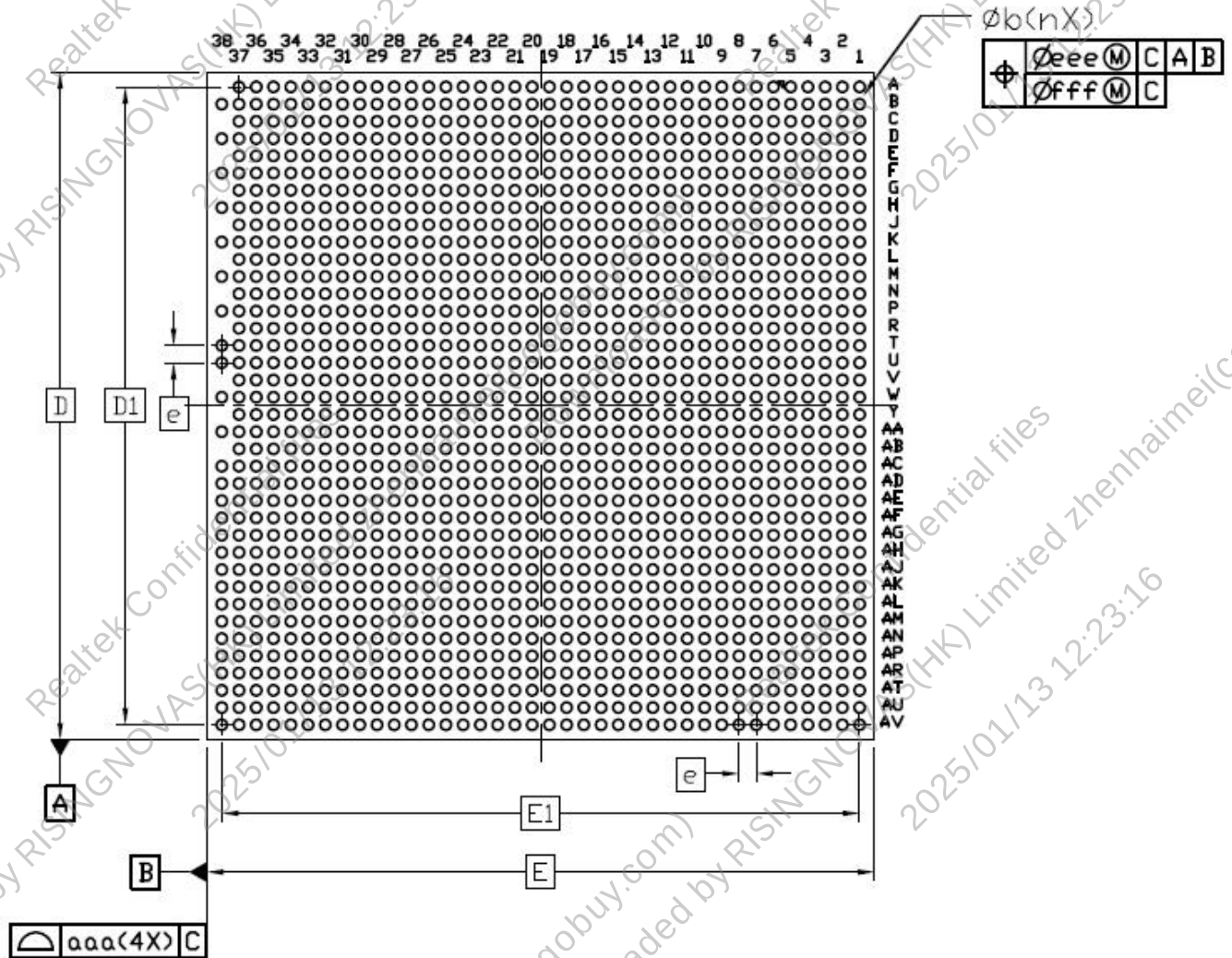
TOP VIEW



SIDE VIEW



BOTTOM VIEW





	SYMBOL	COMMON DIMENSIONS		
		MIN.	NOM.	MAX.
TOTAL THICKNESS	A	2.416	2.546	2.676
STAND OFF	A1	0.300	—	0.500
SUBSTRATE THICKNESS	A2	0.746 REF		
THICKNESS FROM SUBSTRATE SURFACE TO THE HIGHEST POINT	A3	— REF		
BODY SIZE	E	31.000 BSC		
	D	31.000 BSC		
BALL DIAMETER		0.500		
BALL WIDTH	b	0.400	—	0.600
BALL PITCH	e	0.800 BSC		
BALL COUNT	n	1433		
EDGE BALL CENTER TO CENTER	E1	29.600 BSC		
	D1	29.600 BSC		
EXPPOSE DIE SIZE	E2	— BSC		
	D2	— BSC		
PACKAGE EDGE TOLERANCE	aaa	0.200		
SUBSTRATE PARALLELISM	bbb	—		
TOP PARALLELISM	ccc	0.350		
COPLANARITY	ddd	0.200		
BALL OFFSET(PACKAGE)	eee	0.200		
BALL OFFSET (BALL)	fff	0.080		



Realtek Semiconductor Corp.
Headquarters

No. 2, Innovation Road II, Hsinchu Science Park,
Hsinchu 300, Taiwan, R.O.C.

Tel: 886-3-5780211 Fax: 886-3-5776047

www.realtek.com